

# 10

MEASUREMENT AND SPACE

# SURFACE AREA AND VOLUME

The Sydney Opera House was designed by Danish architect Jørn Utzon, with construction taking place between 1959 to 1973 at a cost of $102 million. Its shell design represents a ship under full sail, with each concrete shell being segments of a sphere of radius 75.2 metres. The shells are covered by 1 056 006 white and cream 120 mm square tiles. The Sydney Opera House stands on 1.3 hectares of land, is 183 metres long and 120 metres wide at its widest point.




## Chapter outline

| | Working mathematically | | | | |
|---|---|---|---|---|---|
| **10.01** Metric units | U | F | | R | C |
| **10.02** Precision and absolute error | U | F | | R | C |
| **10.03** Percentage error | U | F | | R | C |
| **10.04** Perimeters and areas of composite shapes | U | F | PS | R | |
| **10.05** Areas of quadrilaterals | U | F | PS | R | |
| **10.06** Circumferences and areas of circular shapes | U | F | PS | R | |
| **10.07** Surface area of a prism | U | F | PS | R | |
| **10.08** Surface area of a cylinder | U | F | PS | R | |
| **10.09** Volumes of prisms and cylinders | U | F | PS | R | C |
| **10.10** Volumes of pyramids and cones* | U | F | PS | R | C |

***PATH**

## Wordbank

**absolute error** The greatest possible error of a measurement, equal to half of the precision (smallest unit) on the measuring instrument's scale

**capacity** The amount of fluid (liquid or gas) in a container

**cross-section** A 'slice' of a solid, taken across the solid rather than along it

**mega- (abbreviated M)** A metric prefix meaning one million

**micro- (abbreviated μ)** A metric prefix meaning one-millionth

**precision** The smallest unit on the scale of a measuring instrument

**prism** A solid shape with identical cross-sections and ends with straight sides

**sector** A region of a circle cut off by 2 radii, shaped like a slice of pizza

**surface area** The total area of all the faces of a solid shape

**Quiz**
Wordbank 10

**U** = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

### Videos (8):

**10.03** Percentage error
**10.06** Circumference of a circle • Area of a circle • Perimeter and area of a sector
**10.07** Surface area of a prism • Surface area of prisms
**10.08** Surface area of a cylinder
**10.09** Volumes of prisms and cylinders

### *Twig* videos (3):

**10.02** Decimal places: Photofinish
**10.06** Beating the U-boats
**10.07** Cubist art

### Quizzes (6):

- Wordbank 10
- SkillCheck 10
- Mental skills 10A
- Mental skills 10B
- Language of maths 10
- Test yourself 10

### Skillsheets (3):

**10.04** What is area?
**10.07** Solid shapes
**10.09** What is volume?

### Worksheets (14):

**10.02** Accuracy in measurement
**10.05** Area ID
**10.06** A page of circular shapes • Applications of area • Back-to-front problems • Area and perimeter investigations • Nets of solids
**10.07** Surface area 1 • Nets of solids
**10.08** Surface area 2 • A page of prisms and cylinders • Applications of area 3
**10.09** A page of prisms and cylinders • Volume and capacity
**10.10** Back-to-front problems (Advanced)
**Mind map:** Surface area and volume (Advanced)

### Puzzles (4):

**10.06** Carpet talk
**10.08** Car song
**10.09** Formula matching game
**Language of maths** Surface area and volume crossword (Advanced)

### Technology (3):

**10.07, 10.08** Surface area calculator
**10.09** Measuring pyramids • Approximating the volume of a cone

### Spreadsheets (1):

**10.07, 10.08** Surface area calculator

### Presentation (1):

**10.07** Surface area

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

## In this chapter you will:

- learn the metric prefixes for very small and very large units of measurement
- calculate the absolute error and percentage error of a measurement
- calculate the perimeters and areas of triangles, quadrilaterals, circles, sectors and composite shapes
- calculate the surface areas, volumes and capacities of right prisms and cylinders
- (PATH) calculate the volumes of pyramids and cones

## SkillCheck ANSWERS ON P. 640

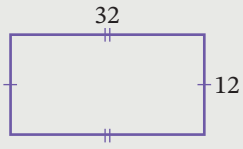
Quiz
SkillCheck 10



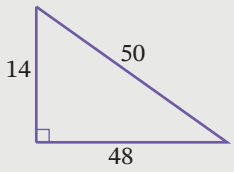
**1** The measurements shown on the shapes below are in centimetres. For each shape, find (correct to one decimal place if necessary):

**i** its perimeter **ii** its area

**a**



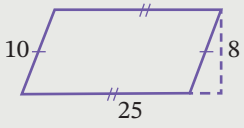

**b**



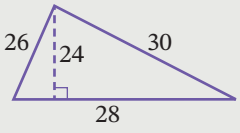

**c**



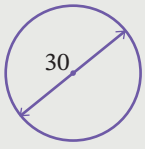

**d**




**e**

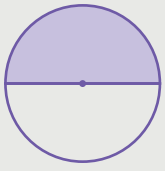



**f**

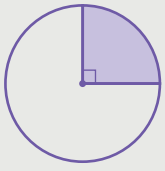



**2** What fraction of each circle is shaded?

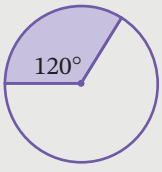
**a**



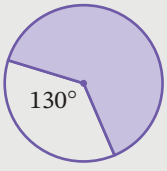
**b**



**c**

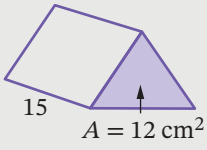



**d**

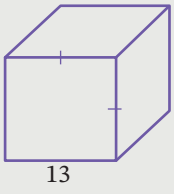



**3** Calculate the volume of each prism (all units are in cm).

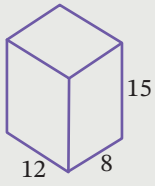
**a**



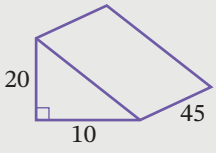

**b**




**c**




**d**



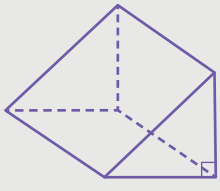

**4** For each prism:

**i** count the number of faces

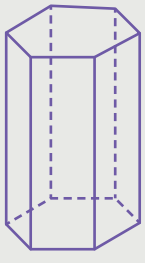
**ii** name the shapes of the different faces

**iii** name the prism.

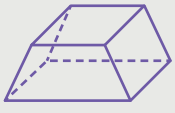
**a**



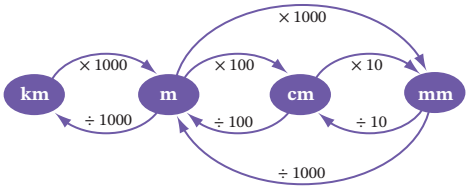
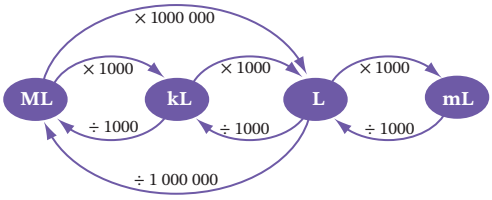
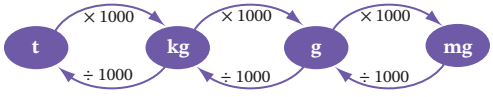
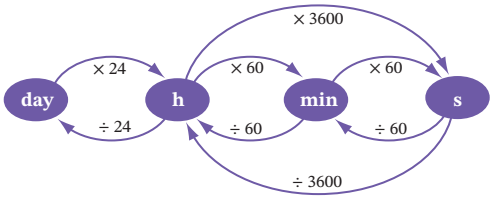
**b**



**c**



# 10.01 Metric units

| Length | Capacity |
|---|---|
| 1 cm = 10 mm<br>1 m = 100 cm = 1000 mm<br>1 km = 1000 m | 1 L = 1000 mL<br>1 kL = 1000 L<br>1 ML = 1000 kL = 1000 000 L |
|  |  |
| **Mass** | **Time** |
| 1 g = 1000 mg<br>1 kg = 1000 g<br>1 t = 1000 kg | 1 min = 60 s<br>1 h = 60 min = 3600 s<br>1 day = 24 h |
|  |  |

Other than for time, units of the metric system have prefixes based on powers of 10.

- **kilo-** means 1000 (thousand) or $10^3$ — 1 kilogram = 1000 grams
- **mega-** means 1 000 000 (million) or $10^6$ — 1 megalitre = 1 000 000 litres
- **centi-** means $\frac{1}{100}$ (hundredth) or $10^{-2}$ — 1 centimetre = $\frac{1}{100}$ metre
- **milli-** means $\frac{1}{1000}$ (thousandth) or $10^{-3}$ — 1 millilitre = $\frac{1}{1000}$ litre

## Example 1

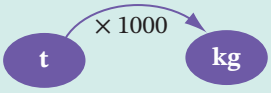
Convert:

**a** 5.8 t to kg **b** 16 km to cm **c** 23 700 mL to L **d** 1500 min to h

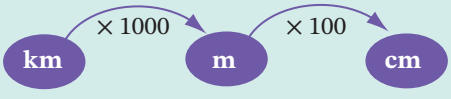
### Solution

**a** $5.8\text{ t} = 5.8 \times 1000\text{ kg}$
$= 5800\text{ kg}$

To convert from large to small units, multiply (×).

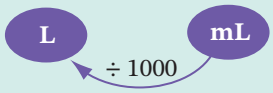



**b** $16\text{ km} = 16 \times 1000 \times 100\text{ cm}$
$= 1\,600\,000\text{ cm}$

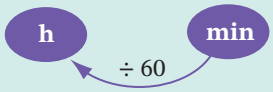


**c** $23\,700\text{ mL} = 23\,700 \div 1000\text{ L}$
$= 23.7\text{ L}$

To convert from small to large units, divide (÷).




**d** $1500\text{ min} = 1500 \div 60\text{ h}$
$= 25\text{ h}$





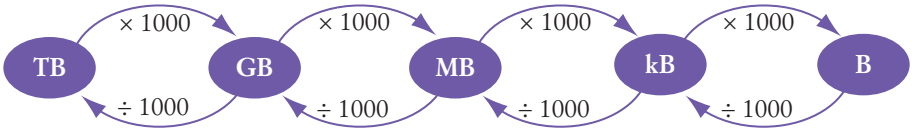

## Metric prefixes

This table shows a more detailed list of metric prefixes.

| Prefix (abbreviation) | Meaning | Example |
|---|---|---|
| pico (p) | $10^{-12}$ | |
| nano (n) | $10^{-9} = \frac{1}{1000000000}$ | nanosecond (ns) = billionth of a second |
| micro (μ) | $10^{-6} = \frac{1}{1000000}$ | microgram (μg) = millionth of a gram |
| milli (m) | $10^{-3} = \frac{1}{1000}$ | millibar (mbar) = thousandth of a bar (air pressure) |
| centi (c) | $10^{-2} = \frac{1}{100}$ | centimetre (cm) = hundredth of a metre |
| deci (d) | $10^{-1} = \frac{1}{10}$ | decibel (dB) = tenth of a bel (sound level) |
| deca (da) | $10^{1} = 10$ | decametre (dam) = ten metres |
| hecto (h) | $10^{2} = 100$ | hectopascal (hPa) = hundred pascals (air pressure) |
| kilo (k) | $10^{3} = 1000$ | kilojoule (kJ) = thousand joules (energy) |
| mega (M) | $10^{6} = 1\,000\,000$ | megahertz (MHz) = million hertz (frequency) |
| giga (G) | $10^{9}$ | gigalitre (GL) = billion litres (water) |
| tera (T) | $10^{12}$ | terawatt (TW) = trillion watts (power) |

Metric prefixes are used in units of digital (computer) memory. A **byte**, abbreviated B, is the amount of memory used to store one character, such as @, g, #, 7 or ?.

| Unit | Relationship | Size |
|---|---|---|
| kilobyte (kB) | 1 kB = 1000 B | About half a page of text. |
| megabyte (MB) | 1 MB = 1000 kB | A million bytes, about the size of a novel. |
| gigabyte (GB) | 1 GB = 1000 MB | A billion bytes, about 500 photos or 7.5 hours of video. |
| terabyte (TB) | 1 TB = 1000 GB | A trillion bytes, about the size of all of the books in a large library. |




Some examples of digital memory sizes:

E-mail (without attachment): 75 kB
Photo: 2 MB
Music file: 4 MB
CD: 750 MB
Streaming a video: 790 MB per hour
DVD: 4.7 GB
Blu-ray disc: 25 GB
Online game: 40 to 300 MB per hour

## Example 2

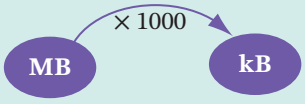
Convert:

**a** 3.2 MB to kB **b** 147 000 MB to GB **c** 2 TB to GB

### Solution

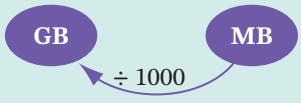
**a** 3.2 MB = 3.2 × 1000 kB
= 3200 kB

To convert from large to small units, multiply (×)

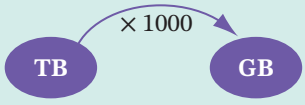



**b** 147 000 MB = 147 000 ÷ 1000 GB
= 147 GB

To convert from small to large units, divide (÷)




**c** 2.6 TB = 2.6 × 1000 GB
= 2600 GB



# Very small and very large units

Some very small and very large units of length, distance and time also use metric prefixes.

**Length and distance**

- One **nanometre** (nm) $= \frac{1}{1\,000\,000\,000}$ m: your fingernails grown 1 nm every second, the diameter of a helium atom is 0.1 nm
- One **micrometre** (μm) $= \frac{1}{1\,000\,000}$ m: used to measure diameters of bacteria
- One **astronomical unit** (AU) = 149 597 871 km: the average distance between the Earth and the Sun, used to measure distances between planets in our solar system
- One **light year** (ly) $= 9.46 \times 10^{12}$ km: the distance light travels in a year, for measuring distances between stars
- One **parsec** (pc) $= 3.0857 \times 10^{13}$ km or 3.26 ly: for measuring distances between stars, the nearest star *Proxima Centauri* is 1.30 parsecs away

**Time**

- One **nanosecond** (ns) $= \frac{1}{1\,000\,000\,000}$ s: the time it takes electricity to travel along a 30 cm wire
- One **microsecond** (μs) $= \frac{1}{1\,000\,000}$ s
- One **millisecond** (ms) $= \frac{1}{1\,000}$ s: the duration of the flash on a camera
- One **decade** = 10 years
- One **century** = 100 years
- One **millennium** = 1000 years
- One **mega-annum** (Ma) = 1 000 000 years: used in geology and paleontology, the *Tyrannosaurus Rex* dinosaur lived 65 Ma ago
- One **giga-annum** (Ga) = 1 000 000 000 years: the Earth formed 4.57 Ga ago

## Example 3

Convert:

**a** 0.73 s to μs

**b** 25 610 000 km to AU

**c** 4.57 Ma to years

**d** 1 920 000 nm to m

### Solution

**a** $0.73 \text{ s} = 0.73 \times 1\,000\,000 \text{ μs}$
$= 730\,000 \text{ μs}$

**b** $25\,610\,000 \text{ km} = 25\,610\,000 \div 149\,597\,871 \text{ AU}$
$= 0.17119...$
$\approx 0.17 \text{ AU}$

**c** $4.57 \text{ Ma} = 4.57 \times 1\,000\,000 \text{ years}$
$= 4\,570\,000 \text{ years}$

**d** $1\,920\,000 \text{ nm} = 1\,920\,000 \div 1\,000\,000\,000 \text{ m}$
$= 1.92 \times 10^{-3} \text{ m}$
$= 0.001\,92 \text{ m}$

**EXERCISE 10.01** ANSWERS ON P. 640

## Metric units U F R C

**1** What metric unit would you use for each measurement? R C

**a** Width of a computer screen
**b** Mass of an apple
**c** Distance between Newcastle and Forster
**d** Capacity of a car's fuel tank
**e** Amount of water in a swimming pool
**f** Time taken to fly overseas

EXAMPLE 1

**2** Convert:

**a** 750 g to mg
**b** 36 kL to L
**c** 67 kg to g
**d** 1.5 L to mL
**e** 6.7 cm to mm
**f** 3.5 km to m
**g** 9.4 t to kg
**h** 3.75 g to mg
**i** 4.25 m to cm
**j** 0.2 kg to g
**k** 26.2 L to mL
**l** 0.3 km to m
**m** 3 kL to mL
**n** 4 days to min
**o** 15 hr to s
**p** 2.7 km to cm
**q** 5 kg to mg
**r** 0.7 m to mm
**s** 29.375 t to g
**t** 18.14 km to mm
**u** 3.27 g to mg

**3** Select the most appropriate answer **A**, **B** or **C** for each measurement.

**a** height of a door
**A** 20 cm **B** 200 cm **C** 20 m

**b** capacity of a soft drink can
**A** 37.5 mL **B** 375 mL **C** 3.5 L

**c** time to run 100 metres
**A** 12.3 s **B** 34.3 s **C** 50.3 s

**d** height of Sydney Tower
**A** 30.5 m **B** 3050 m **C** 305 m

**e** mass of a family car
**A** 100 kg **B** 1000 kg **C** 10 000 kg

**4** Convert:

**a** 8000 L to kL
**b** 90 mm to cm
**c** 180 s to min
**d** 240 min to h
**e** 375 mL to L
**f** 14 700 kg to t
**g** 8600 mg to g
**h** 750 cm to m
**i** 223 mm to cm
**j** 800 g to kg
**k** 125 L to kL
**l** 3.5 kg to t
**m** 17 000 000 mg to kg
**n** 7200 min to days
**o** 45 000 000 g to t
**p** 144 000 s to h
**q** 86 400 s to days
**r** 2500 cm to km
**s** 75 000 mL to kL
**t** 34 700 mm to m
**u** 628 L to ML

**5** The capacity of Sydney Harbour is approximately a quarter of Warragamba Dam's capacity. If the capacity of Warragamba dam is 2013 GL, what is the capacity of Sydney Harbour in litres?

**6** Convert: R C

EXAMPLE 2

| | | |
|---|---|---|
| **a** 4.8 MB to kB | **b** 2.1 TB to GB | **c** 8.5 GB to MB |
| **d** 10.8 MB to B | **e** 2910 B to kB | **f** 740 MB to GB |
| **g** 1050 kB to MB | **h** 5900 GB to TB | **i** 0.94 kB to MB |
| **j** 57 GB to MB | **k** 0.43 GB to kB | **l** 21 TB to GB |

**7** Ritu made a video call lasting $1\frac{1}{4}$ hours. If it used 3MB/min, how much data did the call take?

**8** Calculate the number of MP3 files of average size 5 MB that can be stored on a 64 GB MP3 player. R

**9** Convert, correct to 2 significant figures: R C

EXAMPLE 3

| | | |
|---|---|---|
| **a** 3.5 AU to km | **b** 6000 ms to s | **c** 7.5 centuries to years |
| **d** 12 000 000 nm to m | **e** 0.64 Gm to m | **f** 2.1 ly to km |
| **g** 0.07 s to ms | **h** 0.25 Ma to years | **i** 4 800 000 μs to s |
| **j** 5 290 000 km to Mm | **k** 0.2 pc to km | **l** 6 mm to μm |
| **m** 0.94 Ma to millennia | **n** 849 years to centuries | **o** $8 \times 10^{14}$ km to ly |
| **p** 0.000 008 cm to nm | **q** 20 ly to pc | **r** 156 millennia to Ma |

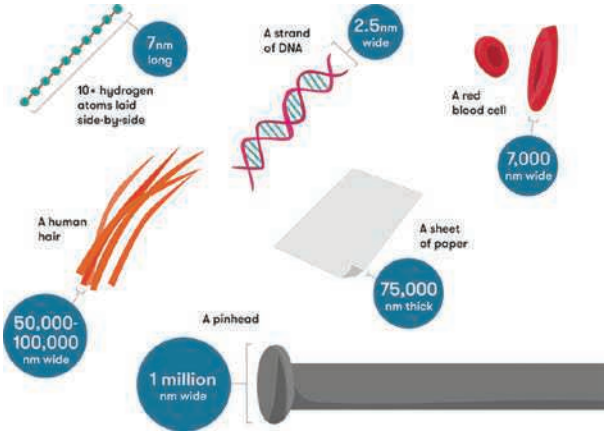
10.01

## Did you know?

### Nanotechnology

One nanometre is equal to one billionth of a metre. This length is so small that it is used to measure cells and bacteria. Nanotechnology is the study of extremely small matter of size 1 to 100 nanometres, also known as the nanoscale. At this scale, matter can possess unusual physical, chemical and biological properties, different to single atom and molecule properties, as nanostructured matter is stronger or contains different magnetic properties in comparison to other forms or sizes of similar material. Such materials can be more chemically reactive, reflect light better or change colour, as their size or form changes.

To understand the comparative size of a nanometre, some examples include:




Reproduced from Nanoscience: thinking big, working small' by Professor Chennupati Jagadish FAA ATSE and Dr Amanda Barnard, Australian Academy of Science, https://www.science.org.au/curious/nanoscience

**Sunscreen, clothing, furniture and computers all incorporate nanotechnology. Research how this is done and find other objects that exist using nanotechnology.**

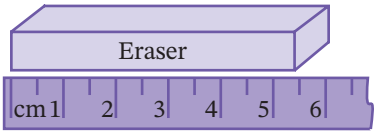
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# 10.02 Precision and absolute error

**Worksheet**
Accuracy in measurement

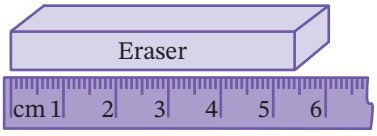
**Video**
Decimal places: Photofinish

All measurements are only approximations.
No measurement is ever **exact**.




For example, we might measure this eraser to be **5.5 cm** long, because the scale on the ruler measures to the nearest 0.5 cm.

If we used a ruler with a more precise scale, such as 0.1 cm markings, then we may find the length to be **5.4 cm**.

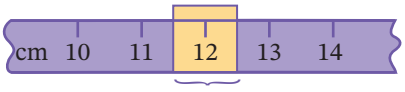



If we used an instrument such as a micrometer that measured to the nearest 0.01 cm, we may find that the length is **5.41 cm**.

Notice that we can always find a more accurate measurement using a more precise measuring instrument. So all measurements are approximations.

A measuring instrument such as a ruler has **limits of accuracy** due to its level of **precision**. 'Accuracy' means how close a measured value is to the true value, while 'precision' means how fine the scale is on the measuring instrument. In measurement, 'error' refers to the difference between a measured value and the true value.

For example, this ruler is marked in centimetres, so any length measured with it can only be given to the nearest centimetre. This ruler has a **precision** of 1 cm.




Any measurement in the shaded region should be recorded as 12 cm. The measured length is 12 cm, but the actual length is $12 \pm 0.5$ cm, meaning '12 centimetres, give or take 0.5 centimetres' or '11.5 to 12.5 cm'.

The **absolute error** of this ruler is **0.5 cm**, which represents the greatest possible error the actual measurement can differ from the recorded measurement (12 cm), the furthest your measurement could be wrong.

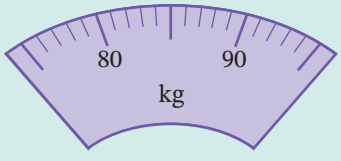
## Absolute error of a measurement

The **absolute error** of a measuring instrument is 0.5 (half) of the precision of the measuring instrument's scale.

## Example 4

Find the absolute error of each measuring scale.

**a**




**b**




### Solution

**a** The precision (size of one unit) of the scale is 1 kg.
The absolute error is $0.5 \times 1 \text{ kg} = 0.5 \text{ kg}$

**b** The precision of the scale is 5 mL.
The absolute error is $0.5 \times 5 \text{ mL} = 2.5 \text{ mL}$.


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**EXERCISE 10.02** ANSWERS ON P. 641

## Precision and absolute error UFRC

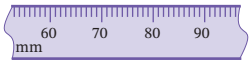
EXAMPLE 4

10.02

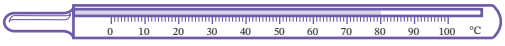
**1** For each measuring instrument, state: R C

**i** its precision **ii** its absolute error

**a**



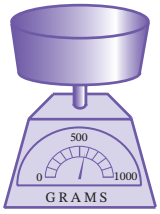

**b**



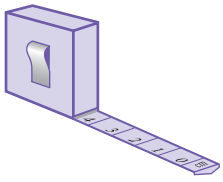

**c**



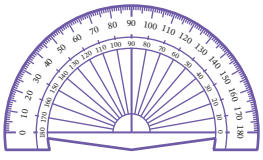

**d**




**e**



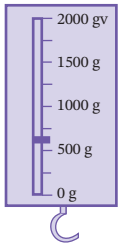
**f**



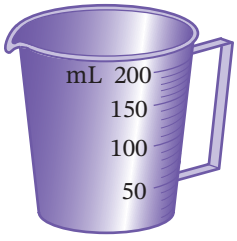
**g**



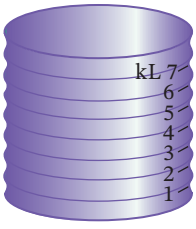

**h**



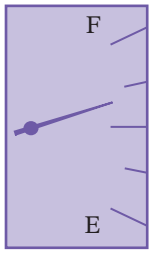

**i**




**j**




**k**



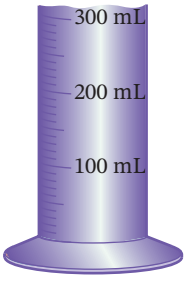

**2** What is the absolute error of this measuring cylinder?
Select the correct answer **A**, **B**, **C** or **D**.

**A** 0.5 mL

**B** 10 mL

**C** 20 mL

**D** 5 mL




**3** Measuring with a ruler, Gemma correctly gives a measurement as 26.5 to 27.5 cm. R C

**a** What is the precision of the ruler?

**b** What is the absolute error of this measurement?

**c** Using a ruler marked in millimetres, Emily gives the same measurement as 26.8 cm. What is the absolute error of this measurement?

**d** What is the range within which this measurement must lie?

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# 10.03 Percentage error

All measurements are approximate. Whether we are using a ruler to measure distance, weighing scales for recipe ingredients or using a thermometer to measure temperature, there is some error in every measurement we make. An absolute error of 1 metre may not be much when measuring the length of a bridge, but it is huge when measuring the length of a bathroom. To understand how big the errors are when something is measured and be able to compare errors in measurement, we can calculate **percentage error**, the absolute error as a percentage of the measurement.

## Percentage error of a measurement

$$\text{Percentage error} = \frac{\text{absolute error}}{\text{measurement}} \times 100\%$$

The smaller the percentage error, the more accurate the measurement, the closer it is to the actual value.

## Example 5

Sydney Tower is 309 metres tall.

Find the percentage error of this measurement, correct to 2 decimal places.



Shutterstock.com/trabantos

### Solution

$\text{Absolute error} = 0.5 \times 1 \text{ m}$
$= 0.5 \text{ m}$

$\text{Percentage error} = \frac{0.5}{309} \times 100$
$= 0.1618123 \ldots$
$\approx 0.16\ \%$

**EXERCISE 10.03** ANSWERS ON P. 641

## Percentage error U F R C

EXAMPLE 5

**1** Copy and complete this table. R C

| | Measurement | Precision | Absolute error |
|---|---|---|---|
| a | 60 cm | | |
| b | 42 seconds | | |
| c | 10.5 mL | | |
| d | 150 cm (to the nearest cm) | | |
| e | 150 cm (to the nearest 10 cm) | | |

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**2** Find the absolute error of each measurement.

**a** 150 mL
**b** 3640 mm
**c** 5.2 g
**d** 200 g (to the nearest g)
**e** 200 g (to the nearest 10 g)
**f** 200 g (to the nearest 100 g)

**3** Dinika's car weighs 1200 kg, to the nearest 100 kg. Calculate the absolute error of this measurement and then its percentage error, correct to 2 decimal places.

EXAMPLE 5

10.03

**4** The Sydney Harbour Bridge spans 503 metres.

**a** Find the absolute error of this measurement.
**b** Hence calculate the percentage error correct to 3 decimal places.

**5** **a** Q1 Tower on the Gold Coast in Queensland is 322.5 metres high. Calculate its percentage error, correct to 3 decimal places.

**b** Which measurement is more accurate, the length of the Sydney Harbour Bridge from Question 4, or the height of the Q1 Tower from part **a**? R C



Shutterstock.com/GiselleA

**6** The longest non-stop passenger flight was from Perth to London in 2022, a distance of 14 498 km. Calculate the percentage error of this measurement, correct to 5 decimal places.

**7** The size of an Olympic swimming pool is 50 metres in length. Given it is measured to the nearest metre, what is the percentage error?

**8** **a** A standard cricket pitch, between the stumps, is 20.12 metres long. Determine the percentage error of this measurement, correct to 3 decimal places.

**b** Which measurement is more accurate, the length of the pool from Question **7**, or the length of the cricket pitch from part **a**? R C

**9** Jenny was filling her car with petrol and told Claire that she had put 44 L of petrol in. Claire looked at the fuel pump and replied that 44.0 L of fuel had been added. Explain which measurement is more accurate. R C

**10** The side length of a cube was measured. Find the percentage error of each measurement.

**a** 4 cm
**b** 4.0 cm

# 10.04 Perimeters and areas of composite shapes

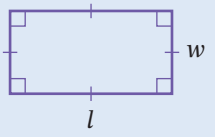
Skillsheet
What is area?

The **perimeter** of a shape is the distance around the shape, the sum of the lengths of the sides of the shape.

The **area** of a shape is the amount of surface covered by the shape, measured in **square units**.

## Areas of rectangles and triangles

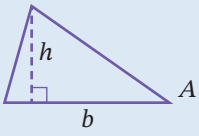
**Rectangle**




$A = \text{length} \times \text{width}$

$A = lw$

**Triangle**




$A = \frac{1}{2} \times \text{base} \times \text{perpendicular height}$

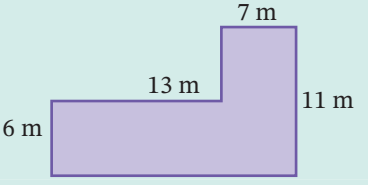
$A = \frac{1}{2}bh$

## Example 6

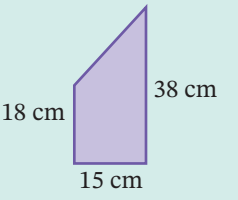
For each composite shape, find:

**i** its perimeter **ii** its area.

**a**

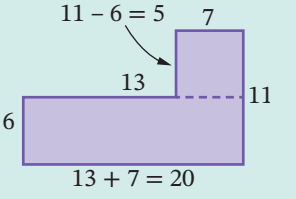



**b**




### Solution

**a i** Find any unknown lengths on the shape.



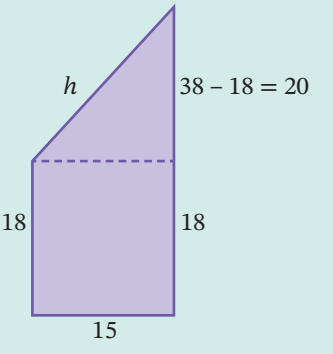

Perimeter $= 6 + 13 + 5 + 7 + 11 + 20$

$= 62$ m

**ii** Area $= (7 \times 5) + (20 \times 6)$ Sum of areas of 2 rectangles

$= 155$ m$^2$

**b i** Divide the shape into a rectangle and a triangle, and find the unknown length, $h$, using Pythagoras' theorem.




$h^2 = 15^2 + 20^2$

$= 625$

$h = \sqrt{625}$

$= 25$ cm

Perimeter $= 18 + 25 + 38 + 15$

$= 96$ cm


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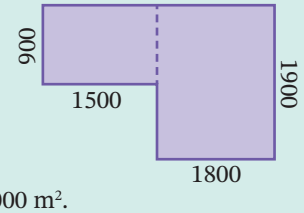
**ii** $\text{Area} = \left(\frac{1}{2} \times 15 \times 20\right) + (15 \times 18)$ Area of triangle + Area of rectangle

$= 420 \text{ cm}^2$

## Example 7

This is the plan of a farm up for sale.

All measurements are in metres.




**a** Find the area of the farm.

**b** Express the area in square kilometres.

**c** Find the area of the property in **hectares**, where 1 ha = 10 000 m².

### Solution

**a** $\text{Area} = 900 \times 1500 + 1900 \times 1800$

$= 4\,770\,000 \text{ m}^2$

**b** Since 1 km = 1000 m, $1 \text{ km}^2 = 1000^2 \text{ m}^2 = 1\,000\,000 \text{ m}^2$

$\therefore \text{Area} = 4\,770\,000 \div 1\,000\,000$

$= 4.77 \text{ km}^2$

**c** 1 ha = 10 000 m²

$\therefore \text{Area} = 4\,770\,000 \div 10\,000$

$= 477 \text{ ha}$

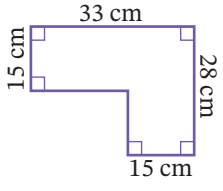
**EXERCISE 10.04** ANSWERS ON P. 641

## Perimeters and areas of composite shapes U F PS R

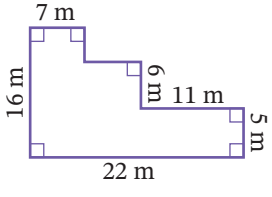
**1** For each composite shape, find: PS R

**i** its perimeter **ii** its area.

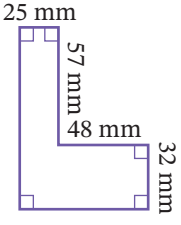
**a**



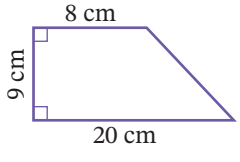

**b**



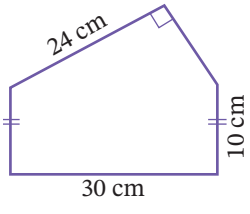

**c**



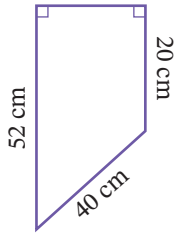

**d**




**e**




**f**






EXAMPLE 6

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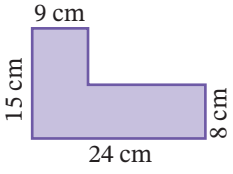
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**2** The length of a rectangular block of land is 112 m and its perimeter is 334 m.
How wide is the block? Select the answer **A**, **B**, **C** or **D**. R

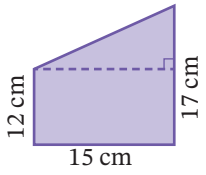
**A** 222 m **B** 110 m **C** 55 m **D** 224 m

**3** Find the area of each composite shape. PS R

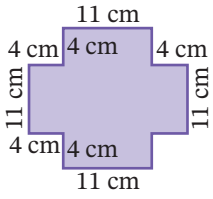
**a**



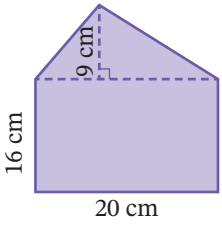

**b**



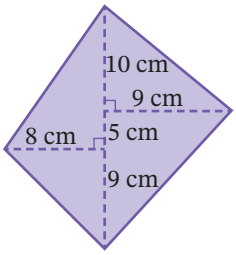

**c**



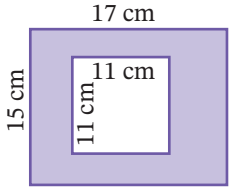

**d**



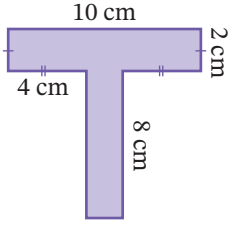

**e**



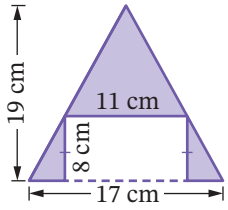

**f**



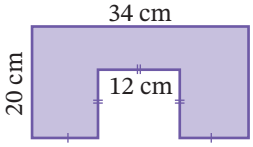

**g**



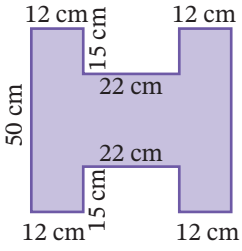

**h**



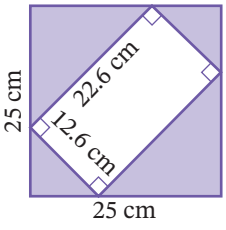

**i**



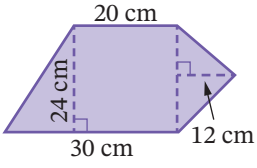

**j**




**k**



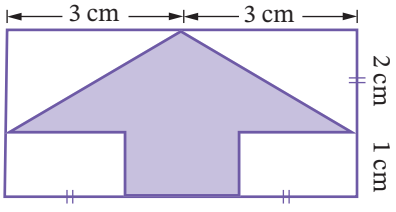

**l**




**4** What is the area of the shaded arrow?
Select the correct answer **A**, **B**, **C** or **D**.

**A** 14 cm$^2$ **B** 8 cm$^2$

**C** 9 cm$^2$ **D** 12 cm$^2$

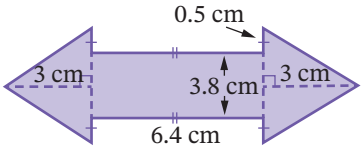



Foundation Standard Complex

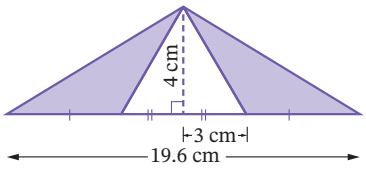

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**5** Find each shaded area. PS R

**a**



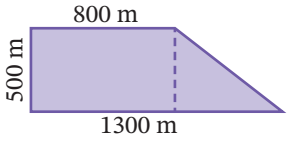

**b**




**6** **a** Find the area of the large block of land shown in the diagram (in $m^2$).

**b** Express the area of the land in:

**i** $km^2$ **ii** hectares




**7** The area of a large rectangular property is 8.6 ha. Calculate the width if its length is 400 m. Select the correct answer **A**, **B**, **C** or **D**.

**A** 21.5 m **B** 215 m **C** 2150 m **D** 0.0215 km

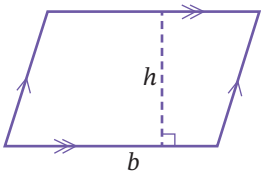
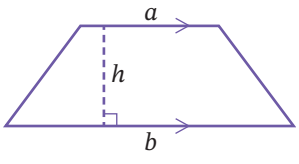
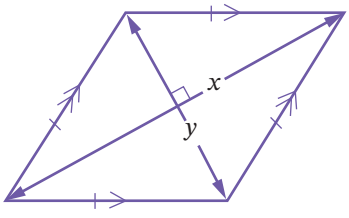
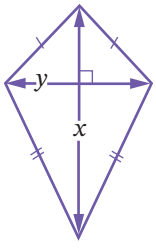
EXAMPLE 7

10.05

# Areas of quadrilaterals 10.05

## Areas of special quadrilaterals

Worksheet
Area ID

| Parallelogram | Trapezium |
|---|---|
|   |   |
| Area = base × perpendicular height<br>$A = bh$ | Area = $\frac{1}{2}$ × (sum of parallel sides) × perpendicular height<br>$A = \frac{1}{2}(a + b)h$ |
| **Rhombus** | **Kite** |
|   |   |

Area = $\frac{1}{2}$ × (product of diagonals)

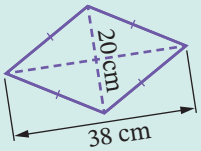
$A = \frac{1}{2}xy$

Foundation Standard Complex

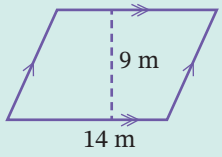
## Example 8

Find the area of each quadrilateral.

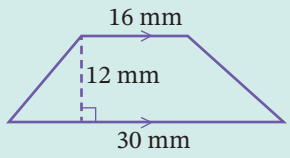
**a**



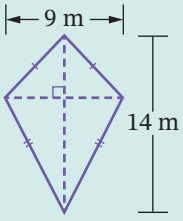

**b**



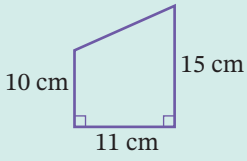

**c**




**d**




**e**




### Solution

**a** Area of rhombus $= \frac{1}{2} \times 38 \times 20$ $\qquad A = \frac{1}{2}xy$

$= 380 \text{ cm}^2$

**b** Area of parallelogram $= 14 \times 9$ $\qquad A = bh$

$= 126 \text{ m}^2$

**c** Area of trapezium $= \frac{1}{2} \times (16 + 30) \times 12$ $\qquad A = \frac{1}{2}(a + b)h$

$= 276 \text{ mm}^2$

**d** Area of kite $= \frac{1}{2} \times 9 \times 14$ $\qquad A = \frac{1}{2}xy$

$= 63 \text{ m}^2$

**e** Area of trapezium $= \frac{1}{2} \times (10 + 15) \times 11$ $\qquad A = \frac{1}{2}(a + b)h$

$= 137.5 \text{ cm}^2$

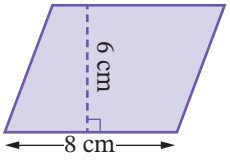
**EXERCISE 10.05** ANSWERS ON P. 641

## Areas of quadrilaterals U F PS R

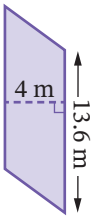
EXAMPLE 8

**1** Find the shaded area of each shape.

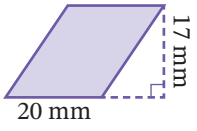
**a**



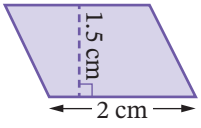

**b**



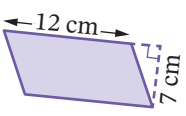

**c**



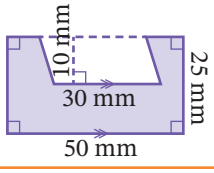

**d**




**e**



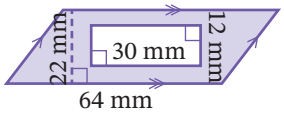

**f**



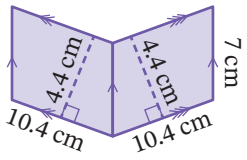

Foundation Standard Complex


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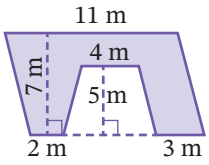
**g**




**h**

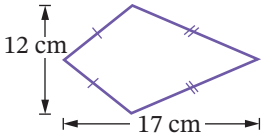



**i**

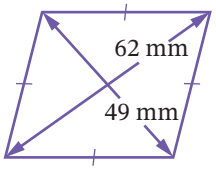



**2** Find the area of each shape.

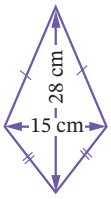
**a**



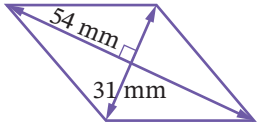

**b**



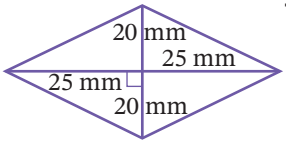

**c**



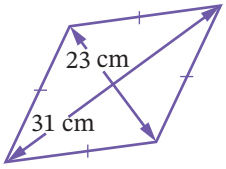

**d**



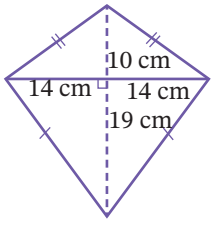

**e**



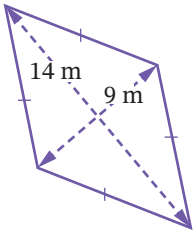

**f**



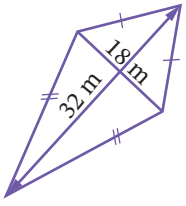

**g**




**h**

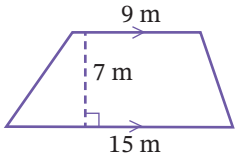



**i**

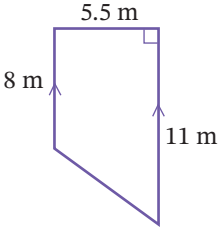



**3** Find the area of each shape.

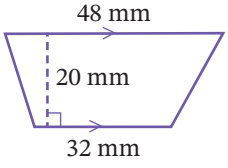
**a**



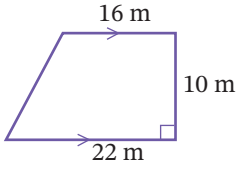

**b**



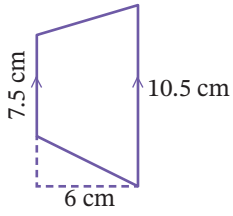

**c**



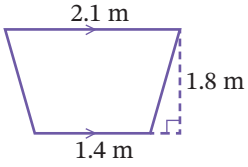

**d**



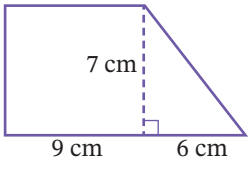

**e**



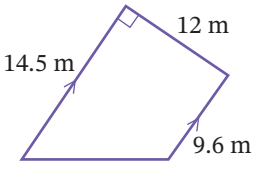

**f**



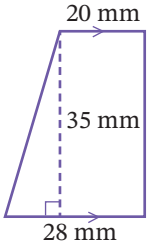

**g**




**h**

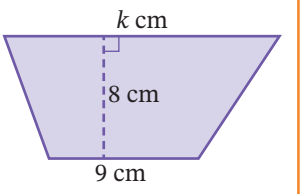



**i**




Foundation | Standard | Complex

**4** The area of the trapezium is 96 cm². What is the length of the missing side? Select the correct answer **A**, **B**, **C** or **D**.

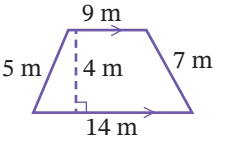



**A** 10 cm **B** 15 cm

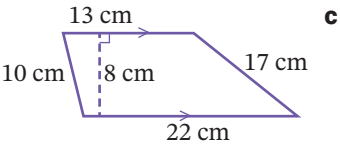
**C** 17 cm **D** 12 cm

**5** Find the area of each shape.

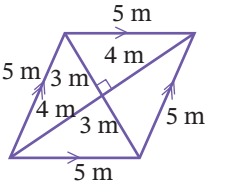
**a**



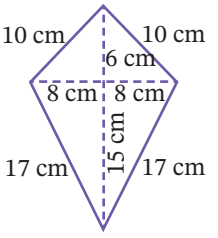

**b**



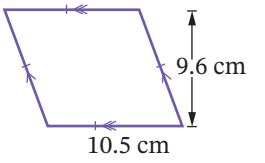

**c**



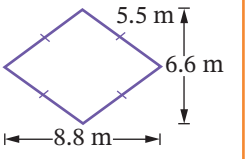

**d**




**e**

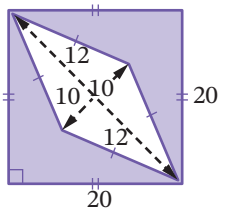



**f**

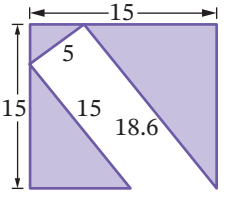



**6** Calculate each shaded area (all measurements are in metres). PS R

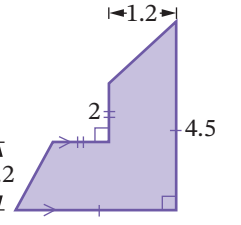
**a**




**b**




**c**




Quiz
Mental skills 10A

## Mental skills 10A: Maths without calculators ANSWERS ON P. 641

### Finding 10%, 20% and 5%

To find 10% or $\frac{1}{10}$ of a number, simply divide the number by 10 by moving the decimal point one place to the left.

**1** Study each example.

**a** $10\% \times 150 = 15\,0. = 15$

**b** $10\% \times \$1256.80 = \$125\,6.\,8 = \$125.68$

**c** $10\% \times 4917 = 491\,7. = 491.7$

**d** $10\% \times \$48.55 = \$4\,8.\,55 = \$4.885$

**2** Now find 10% of each amount.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 190 | **b** | $75 | **c** | 875 | **d** | $202 |
| **e** | $37.60 | **f** | 400 | **g** | $9.25 | **h** | 896 |
| **i** | $2700 | **j** | $3.80 | **k** | $1527.60 | **l** | $72.50 |
| **m** | 3154 | **n** | $10.70 | **o** | 426 | **p** | $24 317.60 |

Foundation Standard Complex


9780170479585

20% is 10% doubled, so to find 20% of a number, first find 10% then double it.

**3** Study each example.

**a**
$20\% \times 700$
$10\% \times 700 = 70$
$\therefore 20\% \times 700 = 70 \times 2$
$= 140$

**b**
$20\% \times \$876$
$10\% \times \$876 = \$87.60$
$\therefore 20\% \times \$876 = \$87.60 \times 2$
$= \$175.20$

**c**
$20\% \times 325$
$10\% \times 325 = 32.5$
$\therefore 20\% \times 325 = 32.5 \times 2$
$= 65$

**d**
$20\% \times \$38.50$
$10\% \times \$38.50 = \$3.85$
$\therefore 20\% \times \$38.50 = \$3.85 \times 2$
$= \$7.70$

**4** Now find 20% of each amount.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 50 | **b** | 620 | **c** | \$2450 | **d** | \$8.60 |
| **e** | 72 | **f** | \$12 700 | **g** | 390 | **h** | \$5.80 |
| **i** | \$45 | **j** | \$84 | **k** | \$4600 | **l** | 320 |

5% is half of 10%, so to find 5% of a number, first find 10% then divide it by 2.

**5** Study each example.

**a**
$5\% \times 180$
$10\% \times 180 = 18$
$\therefore 5\% \times 180 = 18 \div 2$
$= 9$

**b**
$5\% \times \$76$
$10\% \times \$76 = \$7.60$
$\therefore 5\% \times \$76 = \$7.60 \div 2$
$= \$3.80$

**c**
$5\% \times 120$
$10\% \times 120 = 12$
$\therefore 5\% \times 120 = 12 \div 2$
$= 6$

**d**
$5\% \times \$142.20$
$10\% \times \$142.20 = \$14.22$
$\therefore 5\% \times \$142.20 = \$14.22 \div 2$
$= \$7.11$

**6** Now find 5% of each amount.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 2000 | **b** | \$12 | **c** | 50 | **d** | \$27 |
| **e** | \$36.80 | **f** | \$84 | **g** | 800 | **h** | 130 |
| **i** | \$9.60 | **j** | \$138 | **k** | \$72 | **l** | 840 |

# 10.06 Circumferences and areas of circular shapes

The **circumference** of a circle is found by multiplying its **diameter** by **pi** ($\pi$).
The **area** is found by multiplying $\pi$ by the **radius squared.**

**Worksheets**
A page of circular shapes
Applications of area
Back-to-front problems
Area and perimeter investigations

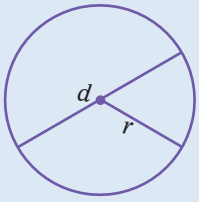
**Puzzle**
Carpet talk

## Circumference of a circle

The **circumference** (**perimeter**) **of a circle** is:

$C = \pi \times \text{diameter}$ or $C = 2 \times \pi \times \text{radius}$

$C = \pi d$ or $C = 2\pi r$




## Area of a circle

The **area of a circle** is:

$$A = \pi \times (\text{radius})^2$$

$$A = \pi r^2$$

$\pi = 3.141\ 592\ 653\ 897\ 93...$ and can be found by pressing the $\pi$ key on your calculator. It has no exact decimal or fraction value, so like a surd such as $\sqrt{2}$, it is an **irrational number**.

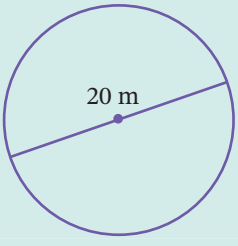
**Videos**
Circumference of a circle
Area of a circle
Beating the U-boats

## Example 9

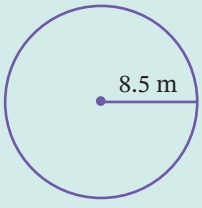
For each circle, calculate correct to 2 decimal places:

**i** its circumference **ii** its area.

**a**




**b**




### Solution

**a i** $C = \pi \times 20$ $C = \pi d$
$= 62.8318...$
$\approx 62.83$ m

**ii** $A = \pi \times 10^2$ $A = \pi r^2$
$= 314.1592...$
$\approx 314.16$ m$^2$

**b i** $C = 2 \times \pi \times 8.5$ $C = 2\pi r$
$= 53.4070...$
$\approx 53.41$ m

**ii** $A = \pi \times 8.5^2$ $A = \pi r^2$
$= 226.9800...$
$\approx 226.98$ m$^2$

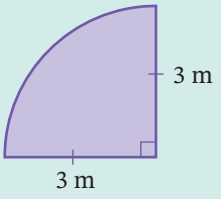
## Example 10

For each sector, calculate correct to one decimal place:

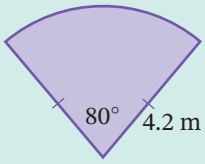
A **sector** is a fraction of a circle 'cut' along 2 radii, like a pizza slice.

**i** its perimeter **ii** its area.

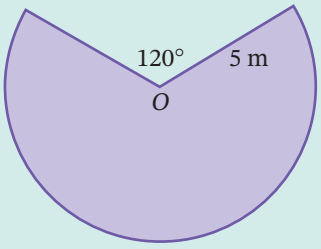
**a**




**b**




**c**




### Solution

**a i** This sector is a **quadrant**, a quarter of a circle.

$\frac{1}{4} \times$ circumference + radius + radius

$$\text{Perimeter} = \frac{1}{4} \times 2 \times \pi \times 3 + 3 + 3$$
$$= 10.71238...$$
$$= 10.7 \text{ m}$$

**ii** $\frac{1}{4} \times$ area of circle

$$\text{Area} = \frac{1}{4} \times \pi \times 3^2$$
$$= 7.06858...$$
$$= 7.1 \text{ m}^2$$

**b i** $\frac{80}{360} \times$ circumference + radius + radius

$$\text{Perimeter} = \frac{80}{360} \times 2 \times \pi \times 4.2 + 4.2 + 4.2$$
$$= 14.26430...$$
$$= 14.3 \text{ m}$$

There are 360° in a circle, but a sector is a fraction of a circle.

**ii** $\frac{80}{360} \times$ area of circle

$$\text{Area} = \frac{80}{360} \times \pi \times 4.2^2$$
$$= 12.31504...$$
$$= 12.3 \text{ m}^2$$

**c i**
$$\text{Sector angle} = 360° - 120°$$
$$= 240°$$
$$\text{Perimeter} = \frac{240}{360} \times 2 \times \pi \times 5 + 5 + 5$$
$$= 30.94395...$$
$$= 30.9 \text{ m}$$

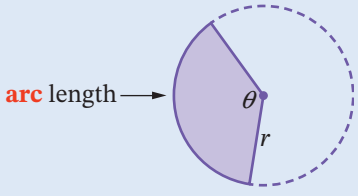
**ii**
$$\text{Area of sector} = \frac{240}{360} \times \pi \times 5^2$$
$$= 52.35987...$$
$$= 52.4 \text{ m}^2$$

## Arc length, perimeter and area of a sector

For a sector with angle θ:

- Arc length $= \frac{\theta}{360} \times 2\pi r$
- Perimeter of the sector $= \frac{\theta}{360} \times 2\pi r + r + r$

  (arc length + radius + radius)
- Area of the sector $= \frac{\theta}{360} \times \pi r^2$



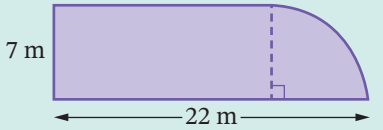

Video
Perimeter and area of a sector

10.06

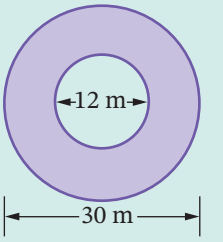
## Example 11

Find, correct to one decimal place, the area of each shape.

**a**




**b**




### Solution

**a** The shape is made up of a rectangle and a quadrant.

Radius of quadrant $= 7$ m

Length of rectangle $= 22 - 7$

$= 15$ m

Area of shape = area of rectangle + quadrant

$= 15 \times 7 + \frac{1}{4} \times \pi \times 7^2$

$= 143.4845...$

$\approx 143.5$ m$^2$

**b** This ring shape is called an **annulus**, it is the area enclosed by 2 circles with the same centre.

Radius of large circle $= \frac{1}{2} \times 30$ m

$= 15$ m

Radius of small circle $= \frac{1}{2} \times 12$ m

$= 6$ m

Shaded area = large circle – small circle

$= \pi \times 15^2 - \pi \times 6^2$

$= 593.7610...$

$\approx 593.8$ m$^2$

**EXERCISE 10.06** ANSWERS ON P. 642

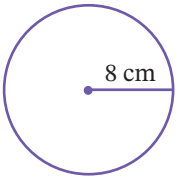
## Circumferences and areas of circular shapes U F PS R

EXAMPLE 9

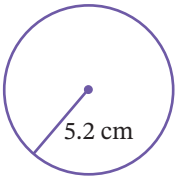
**1** For each circle, calculate correct to 2 decimal places:

**i** its circumference **ii** its area.

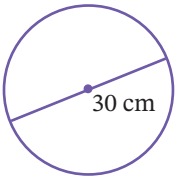
**a**



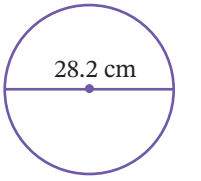

**b**




**c**




**d**

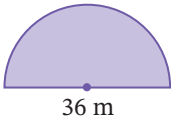



EXAMPLE 10

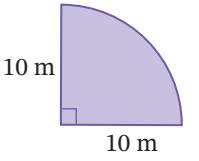
**2** For each sector, calculate correct to one decimal place:

**i** its perimeter **ii** its area.

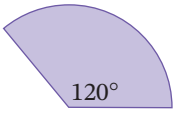
**a**



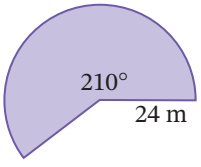

**b**




**c**



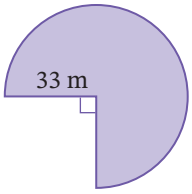

**d**



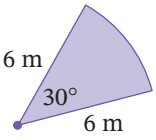

Foundation Standard Complex


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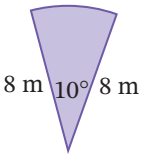
**e**



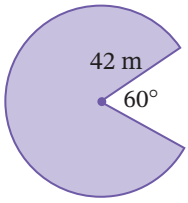

**f**




**g**

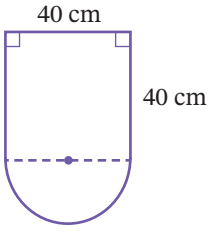



**h**

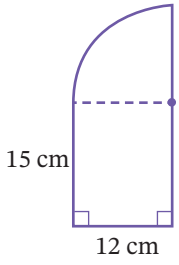



**3** Calculate the perimeter of each shape, correct to 2 decimal places.

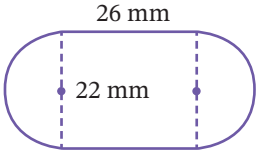
**a**



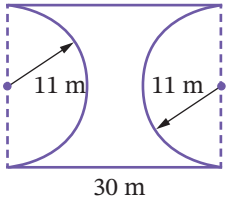

**b**



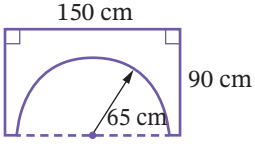

**c**



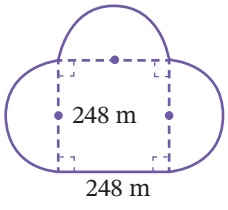

**d**



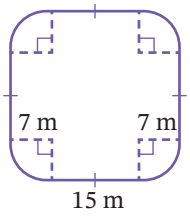

**e**



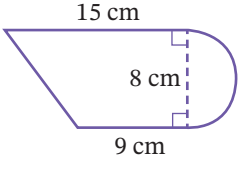

**f**



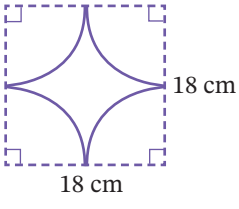

**g**



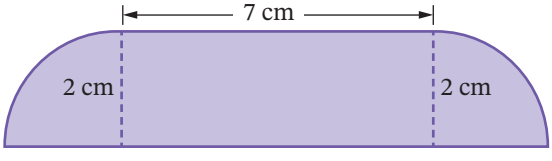

**h**




**i**



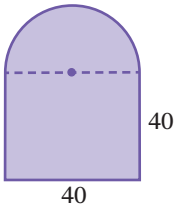

**4** Calculate the area of the shape. Select the correct answer **A**, **B**, **C** or **D**.



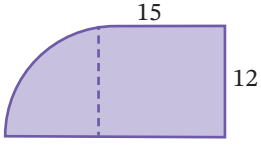

**A** 26.57 cm$^2$ **B** 20.28 cm$^2$ **C** 24.28 cm$^2$ **D** 28.28 cm$^2$

**5** Calculate the area of each shape, correct to the nearest square metre.
All measurements are in metres.

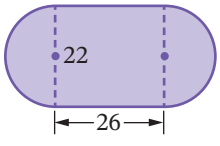
**a**




**b**




**c**

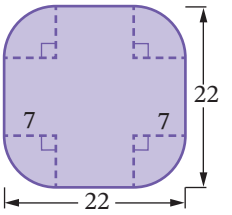



10.06

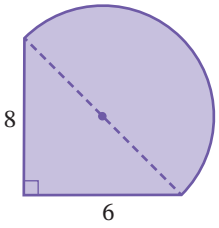
EXAMPLE 11

Foundation Standard Complex

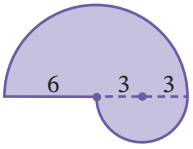
**d**



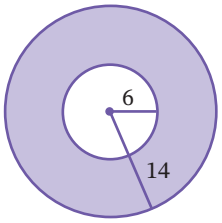

**e**



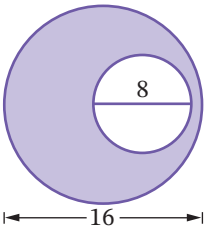

**f**



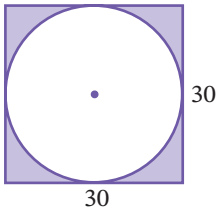

**g**



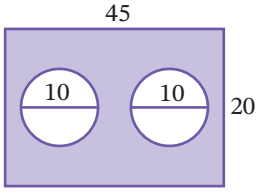

**h**



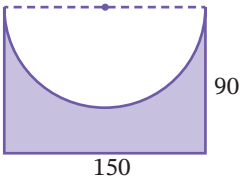

**i**



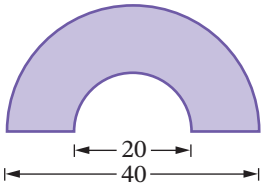

**j**




**k**




**l**




**6** A circular playing field has a radius of 80 m. A rectangular cricket pitch measuring 25 m by 2 m is placed in the middle. The field, excluding the pitch, is to be fertilised. PS R

**a** Calculate, to the nearest square metre, the area to be fertilised.

**b** How much will this cost if the fertiliser is $19.95 per 100 square metres? Give your answer to the nearest dollar.

**7** The diameter of the Earth is 12 756 km.

**a** Find the circumference of the Earth to the nearest kilometre.

**b** The International Space Station orbits the Earth at an average height of 400 km above the Earth's surface. Find the distance it travels in one orbit of the Earth.



Shutterstock.com/Dima Zel

**8** A circular plate of diameter 2 m has 250 holes of diameter 10 cm drilled in it. What is the remaining area of the plate? Answer to the nearest 0.1 $m^2$.

**9** A circular pond of diameter 10 m is surrounded by a path one metre wide. PS R

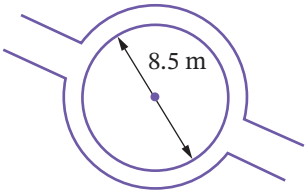
**a** Calculate the area of the path, correct to 2 decimal places.

**b** If pavers are $165 per square metre laid, what is the cost of the path?

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**10** A circular patch of grass has a diameter of 8.5 m. How much further does Arya walk if she walks around the border instead of directly across it? Give your answer correct to the nearest 0.1 m. PS R




**11** A new tractor tyre has a diameter of 120 cm, while a worn tyre has a diameter of 115 cm.

**a** Calculate the difference in circumference between a new and a worn tyre, correct to 3 decimal places.

**b** Over 1000 revolutions, how much further (to the nearest metre) will a new tyre travel compared to a worn tyre?



Shutterstock.com/Anatoliy Kosolapov

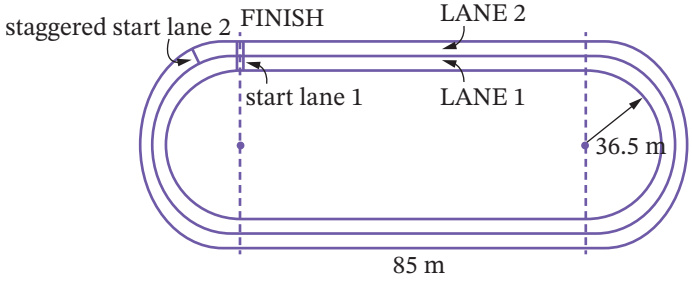
**12** A square courtyard measuring 5 m by 5 m has a semi-circular area added to each side. PS R

**a** Calculate the area of the semi-circular additions, correct to the nearest square metre.

**b** By what percentage has the area of the courtyard increased?

(This can be calculated as $\frac{\text{increase in area}}{\text{original area}} \times 100\%$)

**13** The measurements of a running track are shown below. Each lane is 1 m wide and the athletes run along the insides of the lanes. PS R




**a** Izabella runs one lap in lane 1. What distance does she cover, correct to one decimal place?

**b** Noah runs one lap in lane 2. What distance does he cover, correct to one decimal place?

**c** By how much should the start line be staggered in lane 2 (from lane 1) so that Izabella and Noah run the same distance in one lap?

**14** The wheels of a bicycle have a diameter of 66 cm.

**a** How far (correct to 2 decimal places) will the bicycle move after one rotation of the wheel?

**b** How many rotations of the wheel are needed to cover a distance of 5 km?

# 10.07 Surface area of a prism

The **surface area** of a solid is the total area of all the faces of the solid. To calculate the surface area of a solid, find the area of each face and add the areas together.

**Worksheets**
Surface area 1
Nets of solids

**SkillSheet**
Solid shapes

**Presentation**
Surface area

**Technology**
Surface area calculator

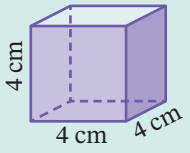
**Spreadsheet**
Surface area calculator

**Video**
Cubist art

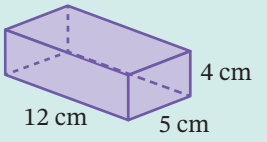
## Example 12

Calculate the surface area of each prism.

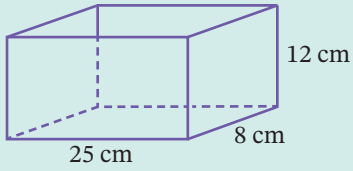
**a** cube




**b** rectangular prism




**c** open rectangular prism




### Solution

**a** A cube has 6 identical faces in the shape of a square. $6 \times$ area of a square

$$\text{Surface area} = 6 \times 4^2$$
$$= 96 \text{ cm}^2$$

**b** A rectangular prism has 6 faces that are not all the same. However, opposite faces, such as the top and bottom, are the same.

$$\text{Surface area} = 2 \text{ top and bottom faces} + 2 \text{ end faces} + 2 \text{ side faces}$$
$$= (2 \times 5 \times 12) + (2 \times 5 \times 4) + (2 \times 12 \times 4)$$
$$= 256 \text{ cm}^2$$

For the surface area of an open solid, we only count the *external* surfaces, not the internal ones (so that we don't count each surface twice).

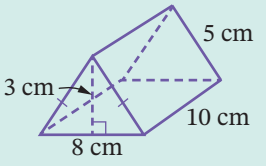
**c** The open rectangular prism has no top so it has 5 faces that are not all the same.

$$\text{Surface area} = \text{Bottom face} + 2 \text{ front and back faces} + 2 \text{ side faces}$$
$$= (25 \times 8) + (2 \times 25 \times 12) + (2 \times 8 \times 12)$$
$$= 992 \text{ cm}^2$$

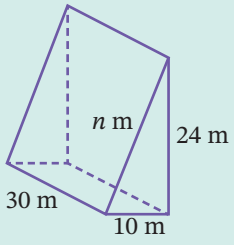
## Example 13

Calculate the surface area of each triangular prism.

**a**




**b**

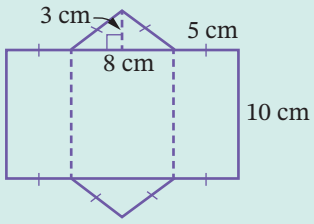



**Videos**
Surface area of a prism
Surface area of prisms


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## Solution

**a** This triangular prism has 5 faces: 2 identical triangles (front and back), 2 identical rectangles (sides) and another rectangle (bottom). Drawing the net of the prism may help you see this.



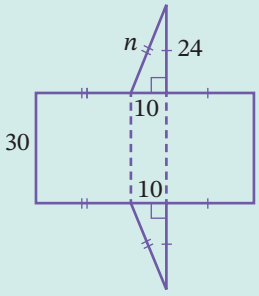

$$\text{Surface area} = (2 \times \tfrac{1}{2} \times 8 \times 3) + (2 \times 10 \times 5) + (8 \times 10)$$
$$= 204 \text{ cm}^2$$

Front and back faces + side faces + bottom face

**b** This triangular prism has 5 faces: 2 identical triangles (front and back), 3 different rectangles.

To calculate the area of the left face, we need to know the value of $n$, which can be found using Pythagoras' theorem.

$$n^2 = 10^2 + 24^2$$
$$= 676$$
$$n = \sqrt{676}$$
$$= 26$$

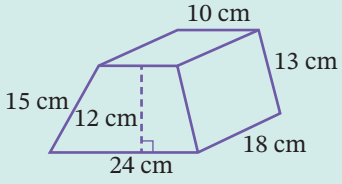



$$\text{Surface area} = (2 \times \tfrac{1}{2} \times 10 \times 24) + (30 \times 26) + (30 \times 10) + (30 \times 24)$$
$$= 2040 \text{ m}^2$$

Front and back + left + bottom + right

# Example 14

Calculate the surface area of this trapezoidal prism.

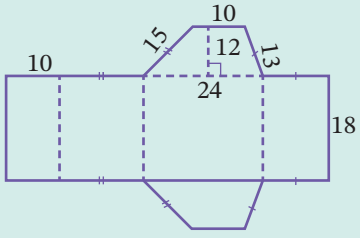



Video
Surface area of a prism

## Solution

This trapezoidal prism has 6 faces: 2 identical trapeziums (front and back), 4 different rectangles.

$$\text{Area of each trapezium} = \tfrac{1}{2} \times (10 + 24) \times 12$$
$$= 204 \text{ cm}^2$$




$$\text{Surface area} = (2 \times 204) + (18 \times 10) + (18 \times 15) + (18 \times 24) + (18 \times 13)$$
$$= 1524 \text{ cm}^2$$

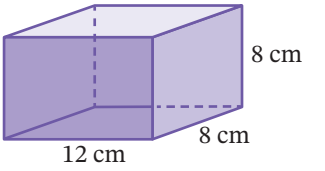
2 trapeziums + 4 rectangles

**EXERCISE 10.07** ANSWERS ON P. 642

## Surface area of a prism U F PS R

EXAMPLE 12

**1** What is the surface area of this rectangular prism?
Select the correct answer **A**, **B**, **C** or **D**.




**A** 320 cm$^2$ **B** 352 cm$^2$

**C** 512 cm$^2$ **D** 768 cm$^2$

**2** Draw a net of each solid.

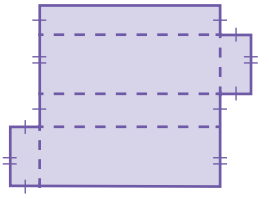
**a** cube **b** rectangular prism **c** triangular prism **d** cylinder

**3** Match each net of a solid to the name of the solid.

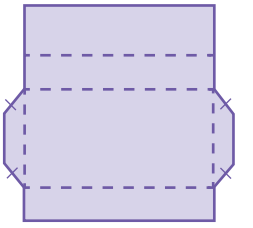
triangular prism cube rectangular prism

square pyramid cylinder trapezoidal prism

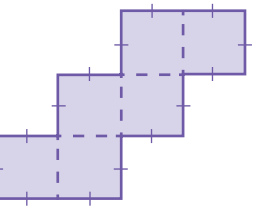
**a**



**b**



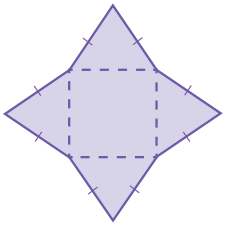
**c**



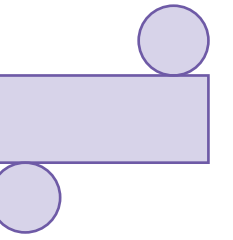
**d**



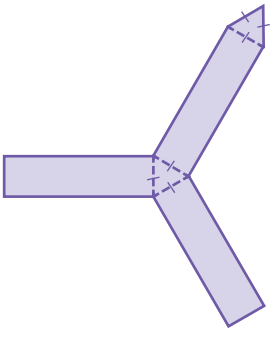
**e**



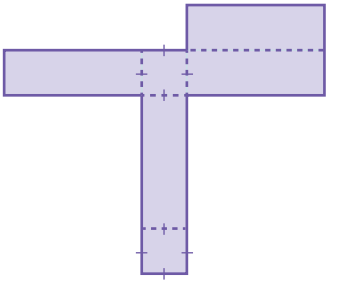
**f**



**g**



**h**



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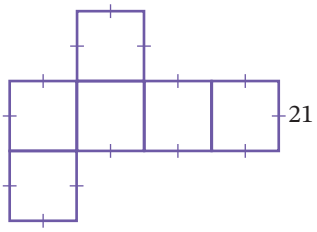

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EXAMPLE 13

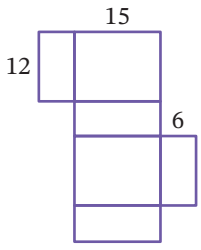
10.07

**4** Name the prism that each net represents, then calculate the surface area of the prism. All lengths are in metres.

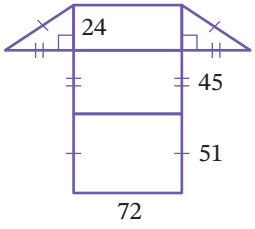
**a**



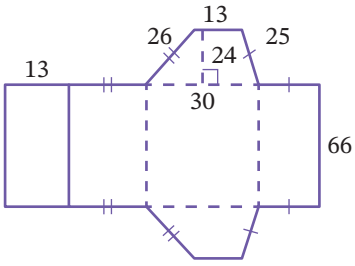

**b**




**c**

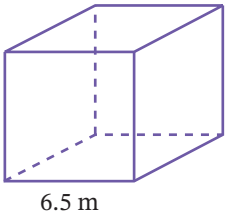



**d**

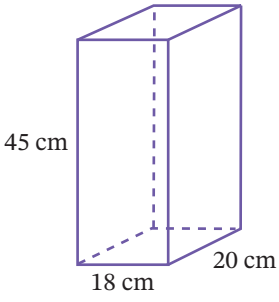



**5** Calculate the surface area of each prism. PS R

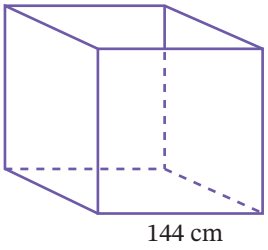
**a** cube



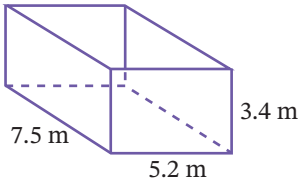

**b** rectangular prism



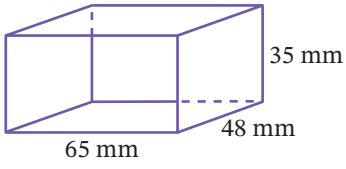

**c** open cube (no top)



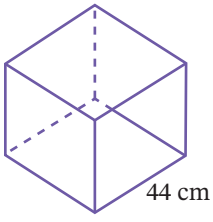

**d** open rectangular prism




**e** rectangular prism, open front and back



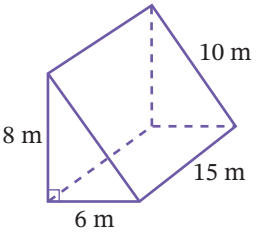

**f** cube, open one end



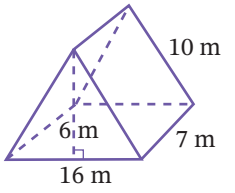

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**6** Find the surface area of each triangular prism. PS R

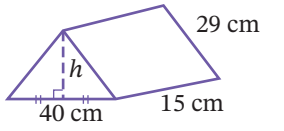
**a**



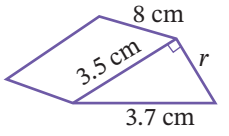

**b**



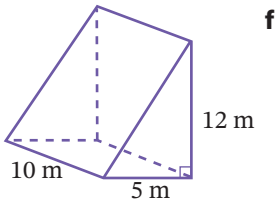

**c**



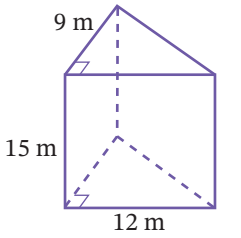

**d**




**e**



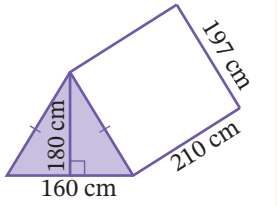

**f**




**7** Yasmin made this tent. PS R

**a** Find the surface area of the tent, including the floor.

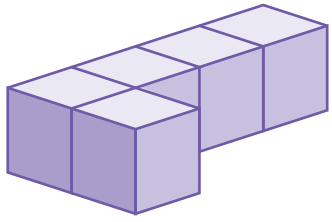
**b** Yasmin bought material at $12 per square metre.
How much did it cost her for the material to make the tent?




**8** 5 cubes of side length 1 cm are joined to make this solid.
What is the surface area of the solid? Select the correct answer **A**, **B**, **C** or **D**.

**A** 17 cm$^2$ **B** 18 cm$^2$

**C** 5 cm$^2$ **D** 22 cm$^2$

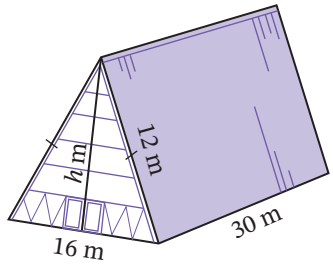


**9** An art gallery has a width of 16 metres, a length of 30 metres and a slant height of 12 metres, as shown in the diagram. PS R

**a** Find the perpendicular height, $h$, of the building, correct to one decimal place.

**b** Find the area of the front wall.

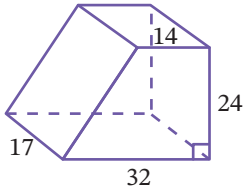
**c** Find the surface area of the art gallery, without the floor.



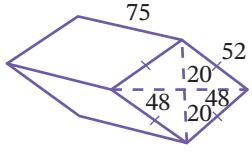

EXAMPLE 14

**10** Calculate the surface area of each prism. Measurements are in centimetres. PS R

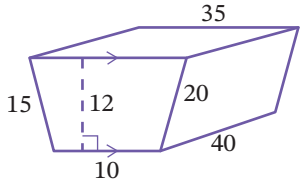
**a**




**b**



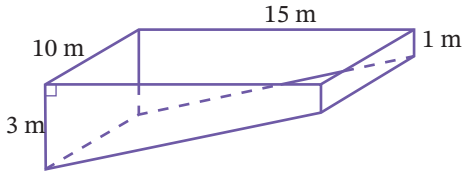

**c**




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**11** This swimming pool is 15 m long and 10 m wide. The depth of the water ranges from 1 m to 3 m. PS R




Calculate, correct to 2 decimal places:

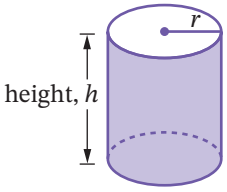
**a** the area of the floor of the pool

**b** the total surface area of the pool.

10.07

## Investigation

### Surface area of a cylinder




Collect 5 different cylindrical cans with paper labels.

**1** Copy this table with rows for Cans 1 to 5 and complete it as you perform the following measurements and calculations.

| | | Circular ends | | | | Curved surface | | | Surface area |
|---|---|---|---|---|---|---|---|---|---|
| | Height | Diameter | Radius | Area | Circumference | Length | Width | Area | |
| Can 1 | | | | | | | | | |
| Can 2 | | | | | | | | | |
| | | | | | | | | | |

**2** Measure the height and diameter of the first can.

**3** Calculate the radius, area and circumference of one circular end, correct to 2 decimal places.

**4** Cut the label off the first can and lay it out flat. What shape is this curved surface?

**5** Measure the length, width and area of the curved surface (label).

**6** Calculate the surface area of the can by adding the areas of the 2 circular ends and the area of the label.

**7** Why is the height of each can and the width of its label the same?

**8** Why is the circumference of each circular end and the length of the label the same?

**9** Repeat Steps **2** to **6** for the other 4 cans.

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# 10.08 Surface area of a cylinder

**Worksheets**
Surface area 2
A page of prisms and cylinders
Applications of area 3

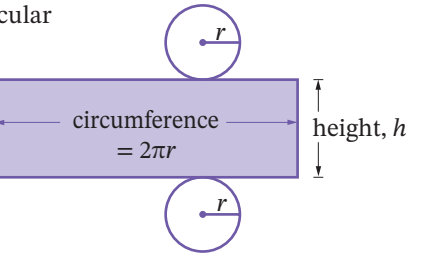
**Puzzle**
Car song

**Technology**
Surface area calculator

**Spreadsheet**
Surface area calculator

**Video**
Surface area of a cylinder

A closed **cylinder** has 3 faces made up of 2 circles (the circular ends) and a rectangle (the curved surface). The length of the rectangle is the circumference of the circular end, while the width of the rectangle is the height of the cylinder.




Surface area of a cylinder = area of 2 circles
+ area of rectangle

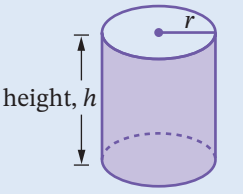
$$SA = 2 \times \pi r^2 + 2\pi r \times h$$
$$= 2\pi r^2 + 2\pi rh$$

## Surface area of a closed cylinder

$SA = 2\pi r^2 + 2\pi rh$

where $r$ = radius of circular base

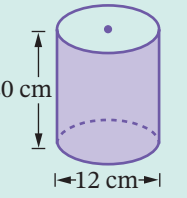
$h$ = perpendicular height




The area of the 2 circular ends = $2\pi r^2$ and the area of the curved surface = $2\pi rh$.

## Example 15

Find, correct to one decimal place, the surface area of a cylinder with diameter 12 cm and height 20 cm.

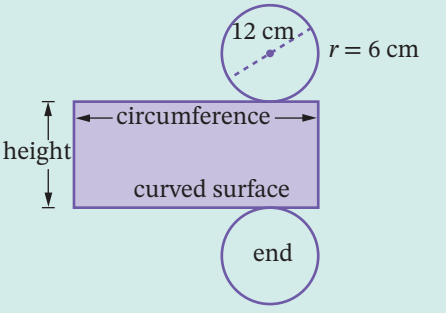



### Solution

Radius = $\frac{1}{2} \times 12$ cm ($\frac{1}{2}$ of diameter)
= 6 cm

Surface area = area of 2 ends + area of the curved surface

$$SA = 2\pi r^2 + 2\pi rh$$
$$= 2 \times \pi \times 6^2 + 2 \times \pi \times 6 \times 20$$
$$= 980.1769...$$
$$\approx 980.2 \text{ cm}^2$$

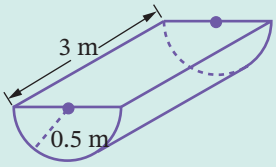




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## Example 16

Find, correct to 2 decimal places, the surface area of this open half-cylinder with radius 0.5 m and height 3 m.




Note: For the surface area of an open solid, we only count the external surfaces, not the internal ones (so that we don't count each surface twice).

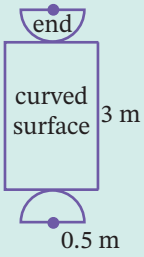
### Solution

Surface area = 2 semicircle ends + $\frac{1}{2}$ × curved surface

$SA = 2 \times (\frac{1}{2} \times \pi \times 0.5^2) + \frac{1}{2} \times (2 \times \pi \times 0.5 \times 3)$

$= 5.49778...$

$\approx 5.50\,\text{cm}^2$




**EXERCISE 10.08** ANSWERS ON P. 642

## Surface area of a cylinder U F PS R

**1** Calculate, correct to 2 decimal places, the surface area of a cylinder with:

**a** radius 7 m, height 10 m

**b** diameter 28 cm, height 15 cm

**c** diameter 6.2 m, height 7.5 m

**d** radius 0.8 m, height 2.35 m

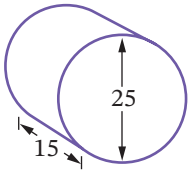
**2** Find, correct to one decimal place, the curved surface area of a cylinder with:

**a** diameter 9 cm, height 32 cm

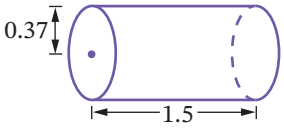
**b** radius 85 mm, height 16 mm

**3** Calculate, correct to one decimal place, the surface area of each solid. All lengths shown are in metres. PS R

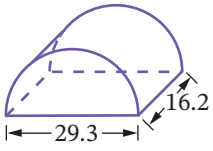
**a** closed cylinder



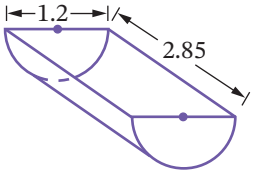

**b** cylinder with one open end



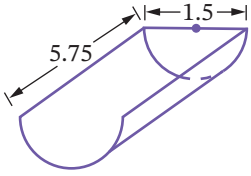

**c** closed half-cylinder



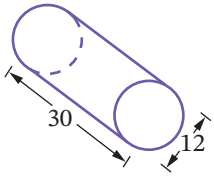

**d** half-cylinder with open top




**e** half-cylinder with open top, one end open




**f** cylinder open at both ends



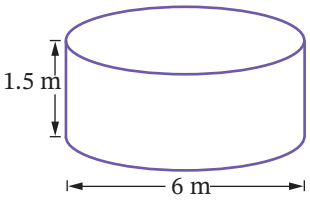

EXAMPLE 15

EXAMPLE 16

10.08

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**4** A swimming pool is in the shape of a cylinder 1.5 m deep and 6 m in diameter. The inside of the pool is to be repainted, including the floor. Find: PS

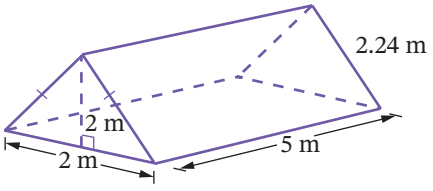



**a** the area to be repainted, correct to one decimal place

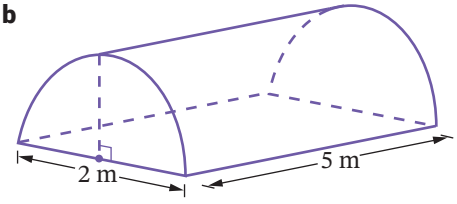
**y** the number of whole litres of paint needed if coverage is 9 $m^2$ per litre.

**5** Which tent has the greater surface area?

**a**

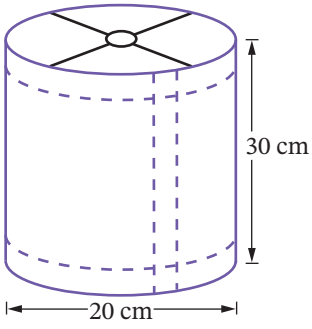



**b**




(*Note*: the floor is included for both tents)

**6** This lampshade is to be covered. Find the area of material needed, correct to one decimal place, if 10% extra is allowed for seams and overlaps. Note: The lampshade is a cylinder open at both ends. PS




# 10.09 Volumes of prisms and cylinders

The **volume** of a solid is the amount of space it takes up. Volume is measured in **cubic units**, for example, cubic metres ($m^3$) or cubic centimetres ($cm^3$).

The **capacity** of a container is the amount of fluid (liquid or gas) it holds, measured in millilitres (mL), litres (L), kilolitres (kL) and megalitres (ML).

**Worksheets**
A page of prisms and cylinders

Volume and capacity

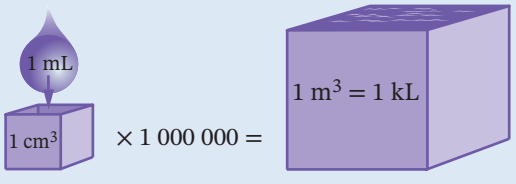
**Skillsheet**
What is volume?

**Puzzle**
Formula matching game

## Volume and capacity

1 $cm^3$ contains 1 mL

1 $m^3$ contains 1000 L or 1 kL




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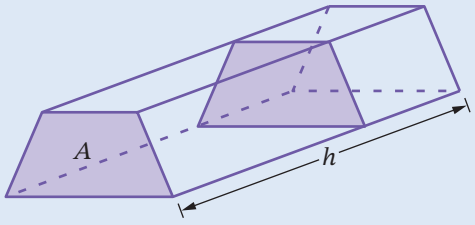

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## Volume of a prism

$V = Ah$

where $A$ = area of base

$h$ = perpendicular height



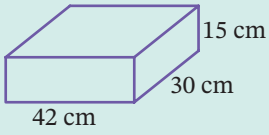

10.09

Video
Volumes of prisms and cylinders

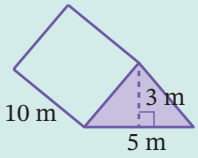
## Example 17

Find the volume of each prism.

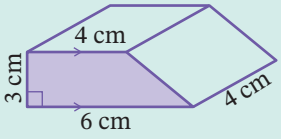
**a**




**b**




**c**




### Solution

**a** $V = 42 \times 30 \times 15$
$= 18\,900 \text{ cm}^3$

For a rectangular prism, volume = length × width × height ($V = lwh$)

**b** $A = \frac{1}{2} \times 5 \times 3$
$= 7.5$

Area of a triangle.

$V = 7.5 \times 10$
$= 75 \text{ m}^3$

$V = Ah$, where height $h = 10$

**c** $A = \frac{1}{2} \times (4 + 6) \times 3$
$= 15 \text{ cm}^2$

Area of a trapezium

$V = 15 \times 4$
$= 60 \text{ cm}^3$

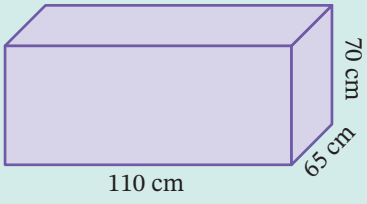
$V = Ah$, where height $h = 4$

## Example 18

A storage box in the shape of a rectangular prism is 110 cm long, 65 cm wide and 70 cm high.

**a** Find the volume of the box in $\text{cm}^3$.

**b** What is the volume in $\text{m}^3$?




### Solution

**a** $V = 110 \times 65 \times 70$
$= 500\,500 \text{ cm}^3$

**b** $1 \text{ m} = 100 \text{ cm}$, so $1 \text{ m}^3 = 100^3 \text{ cm}^3 = 1\,000\,000 \text{ cm}^3$

$\therefore V = 500\,500 \div 1\,000\,000$
$= 0.5005 \text{ m}^3$

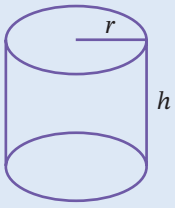
Alternatively, you could convert 110 cm, 65 cm and 70 cm to metres, then multiply them together for the volume.

## Volume of a cylinder

$V = \pi r^2 h$

where $r$ = radius of circular base

$h$ = perpendicular height



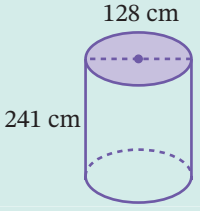

## Example 19

Video
Volumes of prisms and cylinders

For this cylinder, calculate:

**a** its volume, correct to the nearest $cm^3$

**b** its capacity in kL, correct to 1 decimal place.




### Solution

**a** Radius $= \frac{1}{2} \times 128$ cm $= 64$ cm

$\frac{1}{2}$ of diameter

$V = \pi \times 64^2 \times 241$

$= 3\,101\,179.206...$

$\approx 3\,101\,179\ cm^3$

$V = \pi r^2 h$

**b** Capacity $= 3\,101\,179$ mL

$= (3\,101\,179 \div 1000 \div 1000)$ kL

$= 3.101\,179$ kL

$\approx 3.1$ kL

$1\ cm^3 = 1$ mL



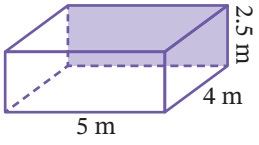

## EXERCISE 10.09 ANSWERS ON P. 642

## Volumes of prisms and cylinders U F PS R C

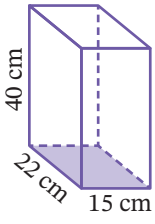
EXAMPLE 17

**1** Find the volume of each rectangular prism using the formula $V = lwh$.

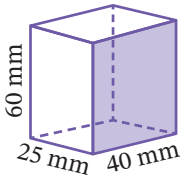
**a**




**b**




**c**




EXAMPLE 18

**2** The volume of a cube is 125 000 $cm^3$. Convert this to $m^3$.
Select the correct answer **A**, **B**, **C** or **D**.

**A** 1250 $m^3$ **B** 12.5 $m^3$ **C** 1.25 $m^3$ **D** 0.125 $m^3$

Foundation Standard Complex

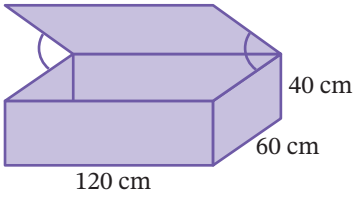

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**3** Find the volume of this toolbox in cubic metres by: PS R C

**a** calculating the volume in $cm^3$, then converting to $m^3$

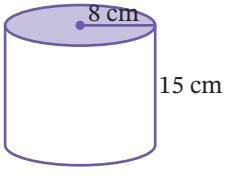
**b** converting each length to metres first, then calculating the volume.

Which method is easier?

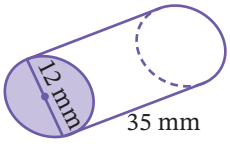



**4** Find, correct to nearest whole unit, the volume of each cylinder.

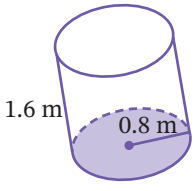
**a**




**b**

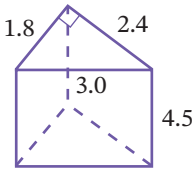



**c**

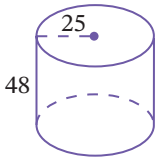



**5** Calculate, correct to one decimal place, the volume of each solid. All lengths are in metres.

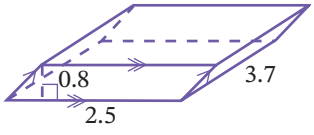
**a**



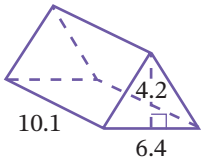

**b**



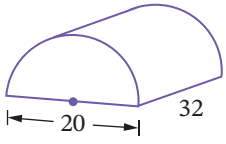

**c**



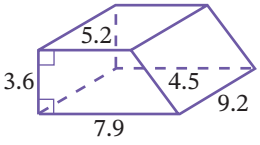

**d**




**e**




**f**

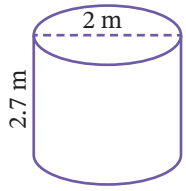



**6** What is the capacity of this cylinder in litres?

Select the closest answer **A**, **B**, **C** or **D**.

**A** 33 900 L **B** 8482 L

**C** 6280 L **D** 16 964 L




**7** A triangular prism has base length 15 cm, height 8 cm and volume 630 $cm^3$. What is its length? R

**8** A fish tank that is 55 cm long, 24 cm wide and 22 cm high is filled to 4 cm below the top. Calculate the amount of water in the tank in litres.

**9** A wedding cake with 3 tiers rests on a table. Each tier is 6 cm high. The layers have radii of 20 cm, 15 cm and 10 cm respectively. Find the total volume of the cake, correct to the nearest $cm^3$.




Shutterstock.com/John Wollwerth

EXAMPLE 19

10.09

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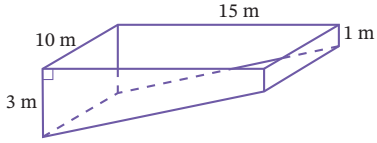
**10** Calculate the volume of the skip bin, correct to the nearest $m^3$.



Shutterstock.com/ Stephen Rees

**11** This swimming pool is 15 m long and 10 m wide. The depth of the water ranges from 1 m to 3 m.

Calculate the capacity of this pool in kilolitres.



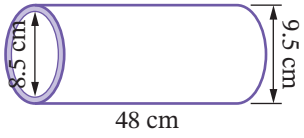

**12** Robyn and Anthony are planning to make a raised vegetable garden. If the dimensions of the garden bed are 7.5 m × 9 m × 60 cm, calculate how much soil they will need to buy. Select the correct answer **A**, **B**, **C** or **D**.

**A** 405 $m^3$ **B** 40.5 $m^3$

**C** 4050 $cm^3$ **D** 4050 $m^3$

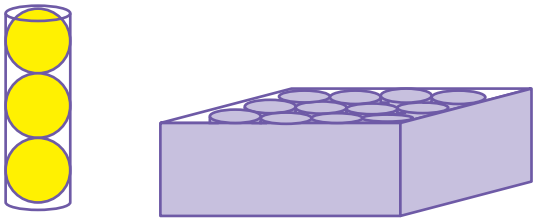

Shutterstock.com/ Alison Hancock

**13** This metal pipe has an inner diameter of 8.5 cm and an outer diameter of 9.5 cm. Calculate, correct to 2 decimal places, the volume of metal needed to make the pipe. PS R




**14** A cube has a volume of 2150 $mm^3$. What is its side length, correct to the nearest millimetre? R

**15** The official diameter of a tennis ball is defined as 6.54 cm to 6.86 cm. PS R C



**a** Find the average diameter of a tennis ball.

**b** The tennis balls are packed in cylinders as shown. Using the larger diameter, calculate the radius and height of one cylinder.

**c** **i** To find the volume of the cylinder, its radius and height are rounded *up* to one decimal place. Explain why, and write the radius and height rounded up.

**ii** Hence, calculate the volume of a cylinder, correct to one decimal place.

**d** For packaging, 12 cylinders are put in a cardboard box, as shown above.

**i** What are the dimensions of the box?

**ii** Calculate the volume of the box.

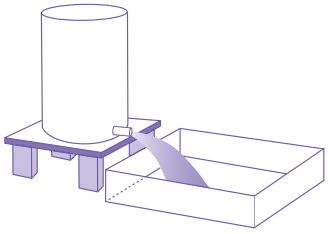
**e** Calculate the volume of the box not taken up by the cylinders.

**f** What percentage (correct to one decimal place) of the box is not taken up by the cylinders?

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**16** A cylindrical water tank of diameter 2 metres and depth 3 metres is full of water. The water flows out of the base of the cylinder into a rectangular prism that is 8 metres long and 4 metres wide. Calculate correct to the nearest centimetre the height of the water in the rectangular prism after all the water has left the cylinder. PS R



10.09

## Investigation

### Volume vs surface area

*You will need:* At least 20 centicubes

**1** Copy the following table.

| Length | Width | Height | Volume | Surface area |
|---|---|---|---|---|
| | | | 8 | |
| | | | 8 | |
| | | | ⋮ | |

**2** Build as many rectangular prisms as you can with a volume of 8 cubes, but with different dimensions. *Note*: $1 \times 1 \times 8$ is the same as $1 \times 8 \times 1$ and $8 \times 1 \times 1$.

**3** Record your dimensions in the table and calculate the surface area of each prism.

**4** What are the dimensions of the solid that has the smallest surface area?

**5** What name do we give to this solid?

**6** Repeat the experiment for a volume of:

**a** 12 cubes **b** 20 cubes.

## Mental skills 10B: Maths without calculators ANSWERS ON P. 643

Quiz
Mental skills 10B

### Finding 15%, $2\frac{1}{2}$%, 25% and $12\frac{1}{2}$%

- To find 10% or $\frac{1}{10}$ of a number, divide by 10.
- To find 5% of a number, find 10% first, then halve it (since 5% is half of 10%).
- So to find 15% of a number, find 10% and 5% of the number separately, then add the answers together.

**1** Study each example.

**a** $15\% \times 80 = (10\% \times 80) + (5\% \times 80)$
$= 8 + 4$
$= 12$

**b** $15\% \times \$170 = (10\% \times \$170) + (5\% \times \$170)$
$= \$17 + \$8.50$
$= \$25.50$

**c** $15\% \times 3600 = (10\% \times 3600) + (5\% \times 3600)$
$= 360 + 180$
$= 540$

**d** $15\% \times \$28 = (10\% \times \$28) + (5\% \times \$28)$
$= \$2.80 + \$1.40$
$= \$4.20$

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**2** Now find 15% of each amount.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 120 | **b** | \$840 | **c** | 260 | **d** | \$202 |
| **e** | \$50 | **f** | 72 | **g** | \$180 | **h** | 400 |
| **i** | \$1600 | **j** | \$22 | **k** | 6000 | **l** | \$350 |

To find $2\frac{1}{2}\%$ of a number, first find 5%, then halve it.

**3** Study each example.

**a** $2\frac{1}{2}\% \times 600$

$10\% \times 600 = 60$

$5\% \times 600 = \frac{1}{2} \times 60 = 30$

$2\frac{1}{2}\% \times 600 = \frac{1}{2} \times 30 = 15$

**b** $2\frac{1}{2}\% \times \$820$

$10\% \times \$820 = \$82$

$5\% \times \$820 = \frac{1}{2} \times 82 = \$41$

$2\frac{1}{2}\% \times \$820 = \frac{1}{2} \times \$41 = \$20.50$

**4** Now find $2\frac{1}{2}\%$ of each amount.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 400 | **b** | 6640 | **c** | \$2000 | **d** | \$880 |
| **e** | 1500 | **f** | \$232 | **g** | 5400 | **h** | \$904 |

To find 25% of a number, halve the number twice as $25\% = \frac{1}{4}$.

**5** Study each example.

**a** $25\% \times 700$

$50\% \times 700 = \frac{1}{2} \times 700 = 350$

$25\% \times 700 = \frac{1}{2} \times 350 = 175$

**b** $25\% \times \$86$

$50\% \times \$86 = \frac{1}{2} \times \$86 = \$43$

$\therefore 25\% \times \$86 = \frac{1}{2} \times \$43 = \$21.50$

**6** Now find 25% of each of each amount.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 2000 | **b** | \$80 | **c** | 18 | **d** | \$25 |
| **e** | \$324 | **f** | \$140 | **g** | 66 | **h** | 298 |
| **i** | \$780 | **j** | \$1700 | **k** | \$126 | **l** | 1160 |

To find $12\frac{1}{2}\%$ of a number, find 25% first, then halve it.
In other words, halve 3 times because $12\frac{1}{2}\% = \frac{1}{8}$.

**7** Study each example.

**a** $12\frac{1}{2}\% \times 400$

$50\% \times 400 = \frac{1}{2} \times 400 = 200$

$25\% \times 400 = \frac{1}{2} \times 200 = 100$

$12\frac{1}{2}\% \times 400 = \frac{1}{2} \times 100 = 50$

**b** $12\frac{1}{2}\% \times \$144$

$50\% \times \$144 = \frac{1}{2} \times \$144 = \$72$

$25\% \times \$144 = \frac{1}{2} \times \$72 = \$36$

$12\frac{1}{2}\% \times \$144 = \frac{1}{2} \times \$36 = \$18$

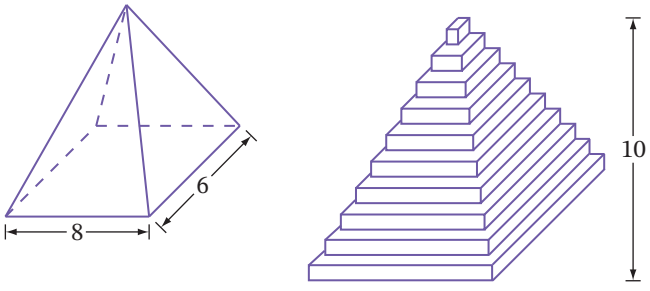
**8** Now find $12\frac{1}{2}\%$ of each amount.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **a** | 1280 | **b** | \$12 | **c** | 60 | **d** | \$260 |
| **e** | \$540 | **f** | \$250 | **g** | 304 | **h** | 1360 |

# Technology

## Approximating the volume of a pyramid

We can approximate the volume of a pyramid by dividing it into of layers of prisms as shown below.




We will create a spreadsheet that approximates the volume of a rectangular pyramid with a base of length 8 units and width 6 units, and a perpendicular height of 10 units.

The volume of each layer can be easily calculated using $V = lwh$. Finding the sum of the layers will then give an approximation of the volume of the pyramid.

Let $n$ be the number of layers. Then the height of each layer is $\frac{10}{n}$.

The length and width of each layer decreases from 8 units and 6 units by a constant amount of $\frac{8}{n}$ and $\frac{6}{n}$ respectively from layer to layer.

**1** Create this spreadsheet for calculating the volue if each layer and the sum of the volumes.

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | | | | Number of layers | | |
| 2 | | | | | | |
| 3 | Height | Length | Width | Thickness of layer | Volume of layer | Sum of volumes |
| 4 | 10 | 8 | 6 | =$A$4/$D$2 | =B4*C4*D4 | =E4 |
| 5 | | =B4-$B4/$D$2 | =C4-$C$4/$D$2 | | | =E5+F4 |
| 6 | | | | | | |
| ⋮ | | | | | | |
| 13 | | | | | | |

**2** **a** To divide the volume of the pyramid into 10 layers, enter **10** in cell D2.

**b** Copy each formula down to row 13.

**c** Explain the results in cells E13 and F13.

**d** How accurate was your result in F13? Explain.

**e** Print out your spreadsheet.

**3** **a** To divide the pyramid into 40 layers to calculate a better approximation, enter **40** in cell D2 and copy each formula down to row 43.

**b** In one or 2 sentences, compare your volume approximation in F43 with the previous approximation in F13.

**4** **a** Enter each of these values in cell D2, copy the formulas down to the appropriate row and write down the approximation for the volume of the pyramid.

**i** 100 (copy down to row 104)

**ii** 200 (copy down to row 204)

**iii** 400 (copy down to row 404)

**b** Use the formula $V = \frac{1}{3}Ah$ to calculate the exact volume of the pyramid.

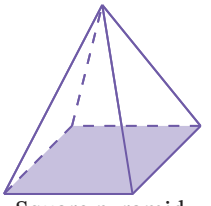
**c** Write a brief report about your results in parts **a** and **b**.

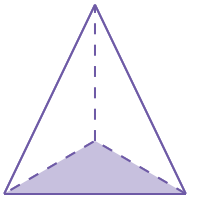
# 10.10 Volumes of pyramids and cones

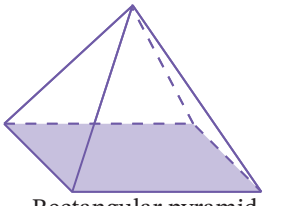
PATH

**Worksheet**
Back-to-front problems (Advanced)

A pyramid is named by the shape of its base.


Square pyramid


Triangular pyramid

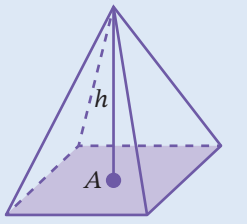

Rectangular pyramid

## Volume of a pyramid

$V = \frac{1}{3}Ah$

where $A$ = area of base

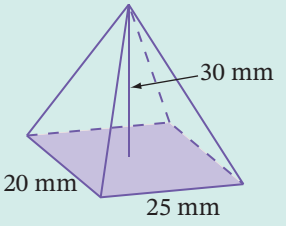
$h$ = perpendicular height



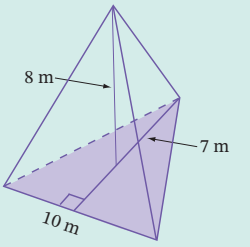

## Example 20

Find the volume of each pyramid.

**a**




**b**





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## Solution

**a** $A = 25 \times 20$ — Area of rectangular base

$= 500 \text{ mm}^2$

$V = \frac{1}{3} \times 500 \times 30$ — $V = \frac{1}{3}Ah$, where height $h = 30$

$= 5000 \text{ mm}^3$

**b** $A = \frac{1}{2} \times 10 \times 7$ — Area of triangular base

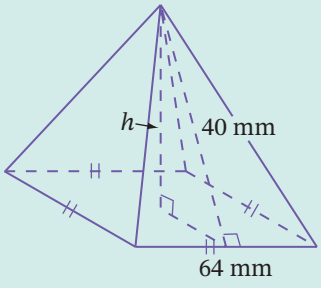
$= 35 \text{ mm}^2$

$V = \frac{1}{3} \times 35 \times 8$ — $V = \frac{1}{3}Ah$, where height $h = 8$

$= 93\frac{1}{3} \text{ m}^3$

## Example 21

Find the capacity (to the nearest millilitre) of a square pyramid with base edge 64 mm and slant height 40 mm.




## Solution

First find $h$, the perpendicular height of the pyramid.

$h^2 = 40^2 - 32^2$ — Using Pythagoras' theorem

$= 576$

$h = \sqrt{576}$

$= 24 \text{ mm}$

$A = 64^2$ — Area of square base

$= 4096 \text{ mm}^2$

$V = \frac{1}{3} \times 4096 \times 24$ — $V = \frac{1}{3}Ah$, where height $h = 24$

$= 32\,768 \text{ mm}^3$

$= 32.768 \text{ cm}^3$ — $1 \text{ cm}^3 = 1000 \text{ mm}^3$

Capacity $= 32.768$ mL — $1 \text{ cm}^3 = 1$ mL

$\approx 33$ mL

PATH

## Example 22

The volume of a square pyramid is 100 cm$^3$. If its height is 12 cm, calculate the length of its base.

### Solution

Let the length of the square base of the pyramid be $x$ cm.

$V = \frac{1}{3}Ah$

$100 = \frac{1}{3} \times x^2 \times 12$

$100 = 4x^2$

$x^2 = \frac{100}{4}$

$= 25$

$x = \sqrt{25}$

$= 5$

$\therefore$ Length of base = 5 cm.

A cone is like a 'circular pyramid' so:

$$\text{Volume} = \frac{1}{3}Ah$$

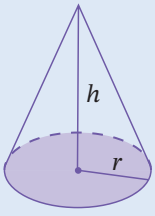
$$V = \frac{1}{3} \times \pi r^2 \times h$$

$$= \frac{1}{3}\pi r^2 h$$

## Volume of a cone

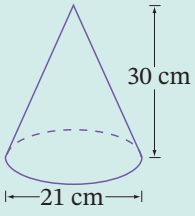
$$V = \frac{1}{3}\pi r^2 h$$

where $r$ = radius of circular base and $h$ = perpendicular height




## Example 23

Calculate the volume of the cone, correct to the nearest cm$^3$.




### Solution

$V = \frac{1}{3} \times \pi \times 10.5^2 \times 30$

$V = \frac{1}{3}\pi r^2 h$, where $r = \frac{1}{2} \times 21 = 10.5$

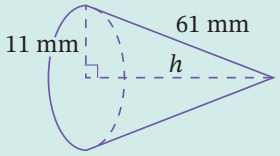
$= 3463.6059...$

$\approx 3464$ cm$^3$


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## Example 24

A cone has a slant edge of 61 mm and a base radius of 11 mm. Find its volume, correct to one decimal place.




### Solution

First, find the height, $h$.

$h^2 = 61^2 - 11^2$
$= 3600$

$h = \sqrt{3600}$
$= 60$

$V = \frac{1}{3} \times \pi \times 11^2 \times 60$
$= 7602.6542...$
$\approx 7602.7 \text{ mm}^2$



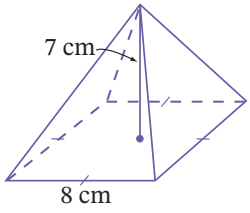
**EXERCISE 10.10** ANSWERS ON P. 643

## Volumes of pyramids and cones U F PS R C

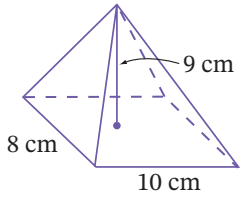
EXAMPLE 20

**1** Calculate the volume of each pyramid (correct to one decimal place where necessary).

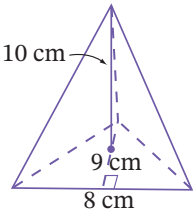
**a**



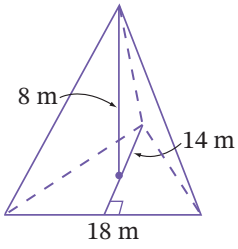

**b**



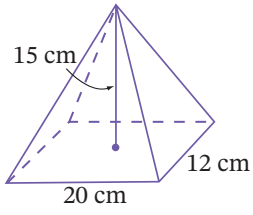

**c**



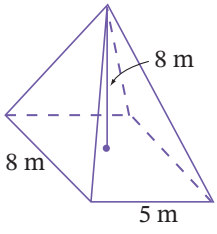

**d**




**e**



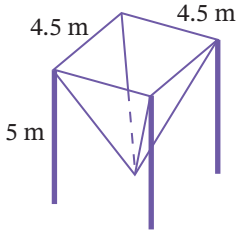

**f**




**2** This grain hopper is in the shape of an inverted square pyramid.

What is its capacity? Select the correct answer **A**, **B**, **C** or **D**.

**A** 15 $m^3$
**B** 33.75 $m^3$
**C** 50.625 $m^3$
**D** 101.25 $m^3$




Foundation | Standard | Complex

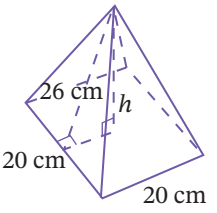
PATH

EXAMPLE 21

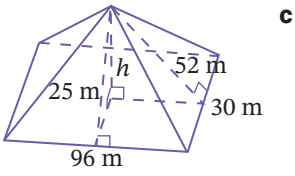
**3** For each pyramid, find correct to 2 decimal places:

**i** its perpendicular height, $h$ **ii** its volume **iii** its capacity.

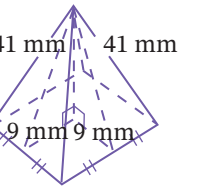
**a**



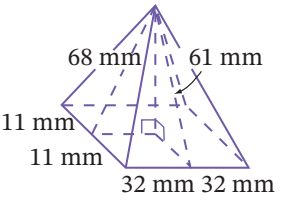

**b**



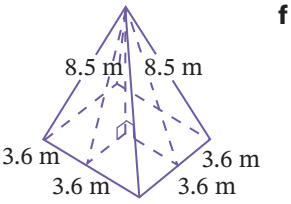

**c**



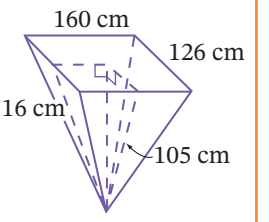

**d**




**e**




**f**




**4** The great pyramid of Khufu (or Cheops) in Egypt was built on a square base with side lengths approximately 230 m.

**a** Find the volume in cubic metres if the original height of the pyramid was 147 m.

**b** There are an estimated 2.3 million stone blocks in the pyramid. Calculate the average volume of each block.




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EXAMPLE 22

**5** The area of the base of a pyramid is 40 m$^2$. If its volume is 360 m$^3$, calculate its perpendicular height. R

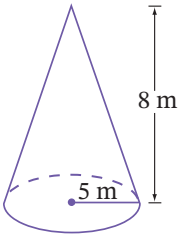
**6** The volume of a square pyramid toy is 1620 mm$^3$. If the length of its base is 8 mm, calculate, correct to the nearest millimetre, the height of the pyramid. R

**7** A square pyramid has a volume of 80 cm$^3$ and a height of 10 cm. Calculate, correct to one decimal place, the length of the base of the pyramid. R

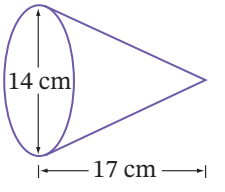
EXAMPLE 23

**8** Calculate the volume of each cone, correct to the nearest whole number.

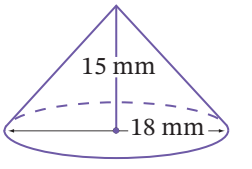
**a**



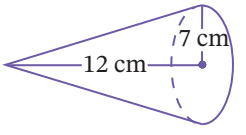

**b**



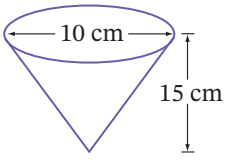

**c**



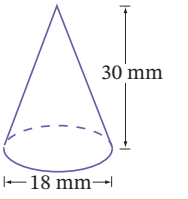

**d**




**e**




**f**




Foundation Standard Complex


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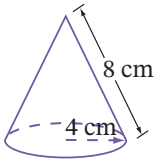
PATH

EXAMPLE 24

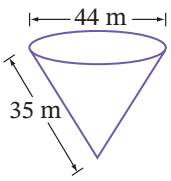
10.10

**9** Calculate the volume of each cone, correct to one decimal place.

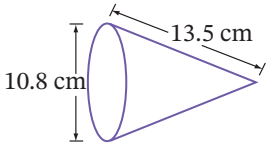
**a**



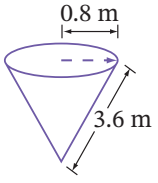

**b**



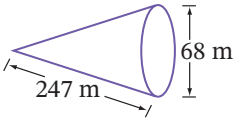

**c**



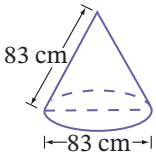

**d**




**e**

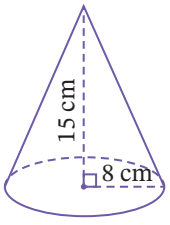



**f**

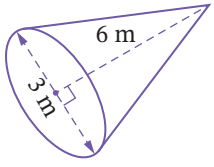



**10** Calculate the volume of each cone, leaving your answer in terms of $\pi$.

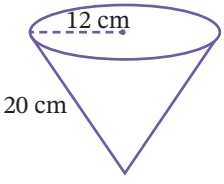
**a**




**b**




**c**

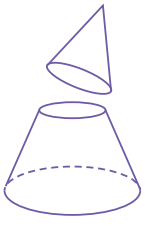



**11** A cone has a volume of 1468 cm$^3$ and a base radius of 12 cm. Find its height, correct to one decimal place. R

**12** A cone has a volume of 820 m$^3$ and a perpendicular height of 10 m. Find its radius, correct to one decimal place. R

**13** A cone has a volume of 150 m$^3$. If the height and radius of the cone are equal in length, calculate the radius of the cone. Give your answer to 2 decimal places. R

**14** A cone has a base radius of 20 cm and a perpendicular height of 48 cm.

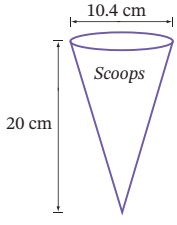


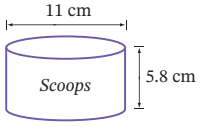
**a** Find the volume of the cone.

**b** The top of the cone is removed at half its height. What percentage of the cone remains? R

**15** *Scoops* popcorn is sold in cones or cylindrical tubs.

Both the cone and the tub of popcorn have the same price. Which container is the better value for money? Give reasons for your answer. PS R C

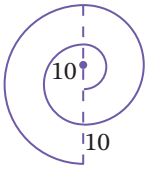



Foundation Standard Complex

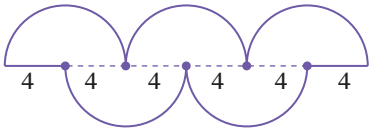
## Power plus ANSWERS ON P. 643

**1** A spiral is formed from 4 semicircles as shown. The diameter of the smallest semicircle is 10 cm, and the semicircles are 10 cm apart. Find the total length of the spiral:




**a** correct to 2 decimal places **b** in terms of $\pi$.

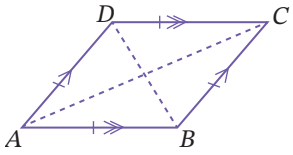
**2** In this pattern, all lengths are in metres. Calculate, correct to 2 decimal places:



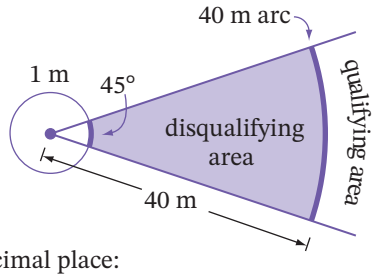

**a** its perimeter

**b** its area

**3** A rhombus has diagonals of length 24 cm and 18 cm. Find its perimeter.




**4** To qualify for the next round of the discus trials, Bronte must throw the discus beyond the 40 m arc. Find, correct to the nearest square metre, the shaded disqualifying area of the sector, given that the small circle has a radius of 1 m.

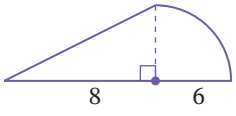



**5** For each composite shape calculate, correct to one decimal place:

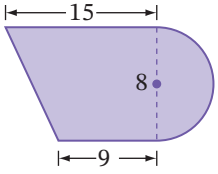
**i** its area **ii** its perimeter.

All measurements are in centimetres.

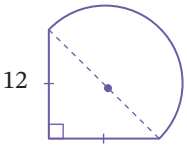
**a**



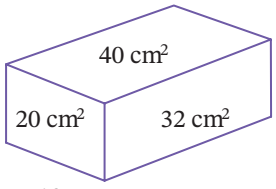

**b**




**c**

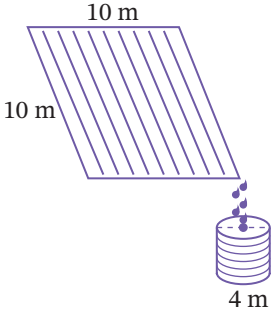



**6** The 3 faces of a rectangular prism have areas as shown. Calculate the volume of the prism.




**7** A cube opened at one end has an external surface area of 1125 cm$^2$. Find its volume.

**8** A 10 m flat square roof drains into a cylindrical rainwater tank with a diameter of 4 m. If 5 mm of rain falls on the roof, by how much (to the nearest mm) does the level of water in the tank rise?




**9** Calculate, correct to one decimal place, the volume of a tetrahedron with a side length of 24 cm.

 9780170479585

# CHAPTER 10 REVIEW



Quiz
Language of maths 10

Puzzle
Surface area and volume crossword

## Language of maths

| | | | |
|---|---|---|---|
| absolute error | arc length | base | capacity |
| circumference | cone | cross-section | curved surface |
| diameter | giga- | micro- (μ) | net |
| percentage error | perpendicular height | pi (π) | precision |
| prefix | pyramid | radius/radii | rectangular prism |
| sector | surface area | triangular prism | |

**1** Draw a circle and label on it:

**a** a sector **b** a diameter **c** an arc length

**2** In the formula $V = Ah$, explain what $V$, $A$ and $h$ stand for.

**3** Which metric prefix means 'one-millionth'?

**4** Write the definition of a **prism** using the word **cross-section**.

**5** How do you find the **surface area** of a solid?

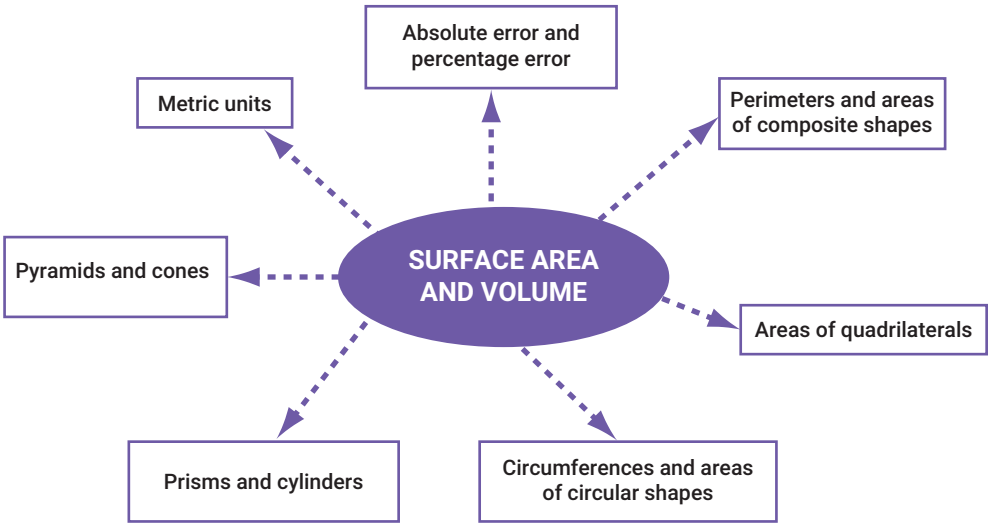
**6** What is the **precision** of your ruler?

## Topic summary

Worksheet
Mind map: Surface area and volume

- Write 10 questions (with solutions) that could be used in a test for this chapter. Include some questions that you have found difficult to answer.
- Swap your questions with another student and check their solutions against yours.
- List the sections of work in this chapter that you did not understand. Follow up this work with your friend or teacher.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



Quiz
Test yourself 10

# TEST YOURSELF 10

ANSWERS ON P. 643

10.01

**1** Convert:

| | | | | | |
|---|---|---|---|---|---|
| **a** | 8 km to m | **b** | 15 min to s | **c** | 0.7 m to mm |
| **d** | 250 000 g to kg | **e** | 8.4 GB to MB | **f** | 9.5 t to kg |
| **g** | 300 min to h | **h** | 34 000 B to kB | **i** | 125 kL to L |
| **j** | 2500 cm to km | **k** | 17 000 000 kg to t | **l** | 8900 GB to TB |
| **m** | 10 500 ms to s | **n** | 9.1 millennia to years | **o** | 2.6 Mm to m |
| **p** | 3 ly to km | **q** | 0.000 000 4 s to μs | **r** | 75 000 000 μm to m |

10.02

**2** For each measuring instrument, state:

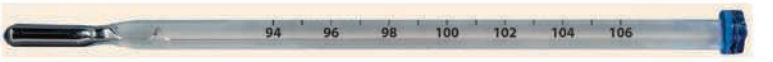
**i** its precision **ii** its absolute error

**a**




Shutterstock.com/DG-Studio

**b**

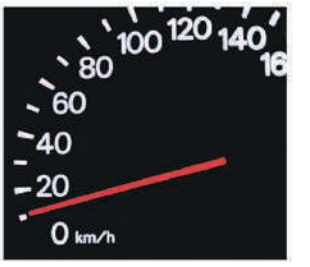



**c**



iStock.com/SergeyMelnikov

**d**




10.03

**3** A super yacht is 24 m long. Calculate:

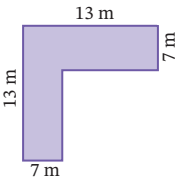
**a** its absolute error

**b** its percentage error correct to 2 decimal places

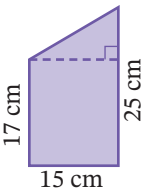
10.04

**4** Calculate the perimeter of each shape.

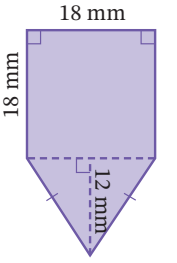
**a**



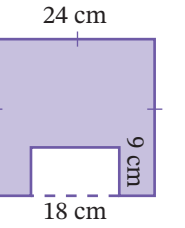

**b**




**c**




**d**



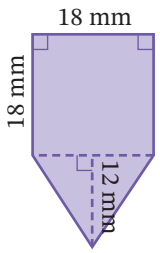

Foundation Standard Complex



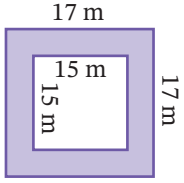
9780170479585

**5** Calculate the area of each shape. 10.04

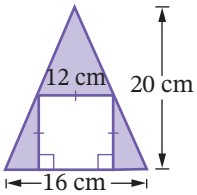
**a**



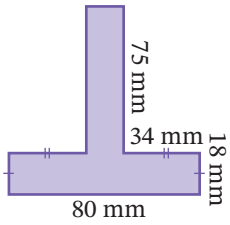

**b**




**c**

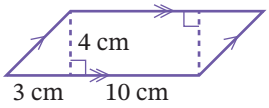



**d**

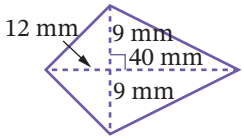



**6** Calculate the area of each quadrilateral. 10.05

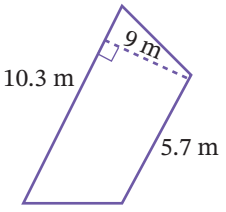
**a**



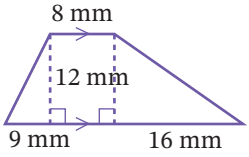

**b**



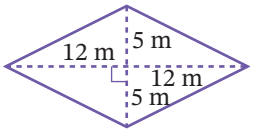

**c**



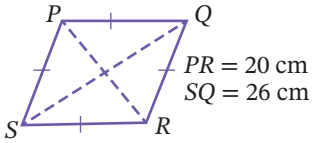

**d**




**e**



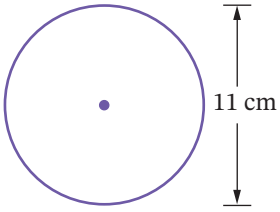

**f**



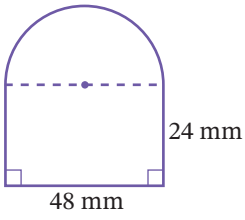

**7** For each shape calculate, correct to one decimal place: 10.06

**i** its perimeter **ii** its area.

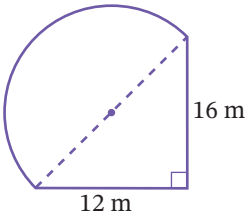
**a**



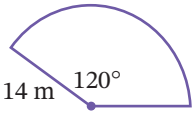

**b**



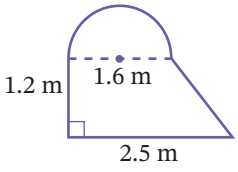

**c**



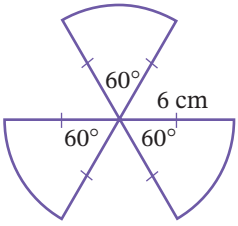

**d**




**e**

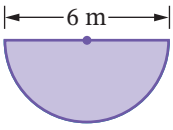



**f**

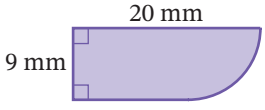



**8** Calculate, correct to 2 decimal places, the area of each shape. 10.06

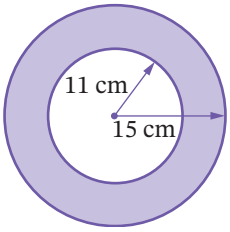
**a**




**b**



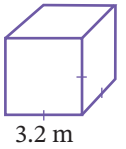

**c**



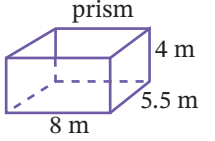

Foundation Standard Complex

10.07 **9** Calculate the surface area of each prism.

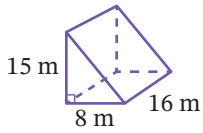
**a** closed cube



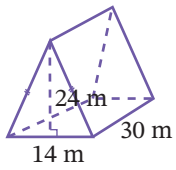

**b** open rectangular prism




**c**



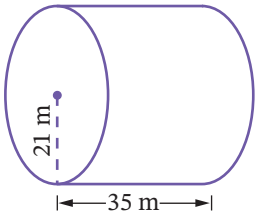

**d**



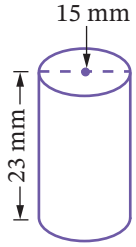

10.09 **10** Calculate the volume of each prism above.

10.08 **11** Calculate, correct to 2 decimal places, the surface area of each cylinder.

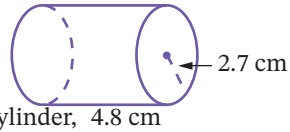
**a**




**b**




**c**



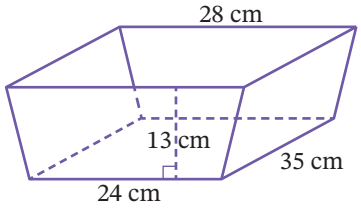

Cylinder, open at one end, 4.8 cm

10.09 **12** Calculate, correct to 2 decimal places, the volume of each cylinder above.

10.08 **13** A cylindrical pipe (open at both ends) is 50 cm long and has a diameter of 15 cm. Calculate its external surface area correct to the nearest square centimetre.

10.09 **14** A rectangular fish tank that is 75 cm long by 55 cm wide by 35 cm deep is filled to 5 cm from the top. How many litres of water is in the tank?

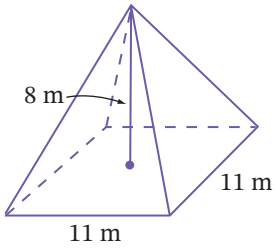
10.09 **15** Calculate the volume of this lunchbox.



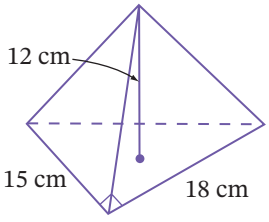

PATH

10.10 **16** Find the volume of each pyramid.

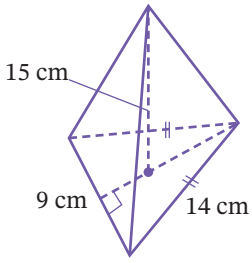
**a**




**b**




**c**

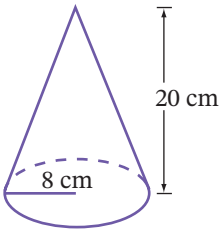



Foundation Standard Complex

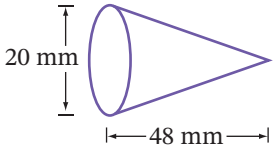

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**17** Find, correct to one decimal place, the volume of each cone.

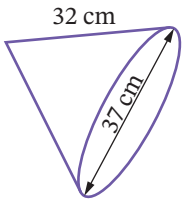
**a**




**b**




**c**




**18** A cone has a volume of 115 $cm^3$ and a base radius of 3.2 cm. Find, correct to one decimal place, the height of the cone.

PATH

10.10

10.10

TEST YOURSELF 10

Foundation | Standard | Complex

# PRACTICE SET 3

ANSWERS ON P. 643

10.01

**1** Copy and complete each conversion.

**a** 750 min = _____ h

**b** 1350 mL = _____ L

**c** 512 MB = _____ GB

**d** 3.005 kg = _____ g

**e** 2.7 km = _____ m

**f** 0.25 MB = _____ kB

8.03

**2** Mason is paid $3.94 for each toy he assembles.

**a** How much does he earn for assembling 140 toys?

**b** How many toys did Mason assemble if he earned $1260.80?

8.04

**3** Katrina's annual salary is $85 000 and she earns another $4800 from investments. Katrina's work-related tax deductions amount to $2380. Calculate Katrina's:

**a** taxable income

**b** income tax payable using the table below.

| Taxable income | Tax on this income |
|---|---|
| 0–$18 200 | Nil |
| $18 201–$45 000 | 19% for each $1 over $18 200 |
| $45 001–$120 000 | $5092 plus 32.5c for each $1 over $45 000 |
| $120 001–$180 000 | $29 467 plus 37c for each $1 over $120 000 |
| $180 001 and over | $51 667 plus 45c for each $1 over $180 000 |

9.02

**4** The shoe sizes of 30 Year 9 students are:

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 6 | 10 | 7 | 6 | 8 | 6 | 7 | 7 | 6 | 5 |
| 5 | 6 | 5 | 8 | 10 | 11 | 9 | 8 | 9 | 5 |
| 6 | 7 | 9 | 8 | 7 | 6 | 5 | 6 | 7 | 8 |

**a** Arrange the data in a frequency table.

**b** Draw a frequency histogram and polygon of this data.

**c** What percentage (correct to one decimal place) of students had shoe sizes less than 8?

**d** Find the mode and the range.

**e** Find the median.

**f** Calculate the mean, correct to one decimal place.

8.01

**5** For a salary of $89 760 p.a., calculate, correct to the nearest cent, the:

**a** weekly income **b** fortnightly income **c** monthly income.

8.03

**6** Bhavin works 38 hours a week and is paid $43.50 per hour. Calculate:

**a** Bhavin's weekly income

**b** the holiday loading of $17\frac{1}{2}$ % on 4 weeks' pay

**c** Bhavin's fortnightly pay when he receives his holiday loading.

Foundation Standard Complex



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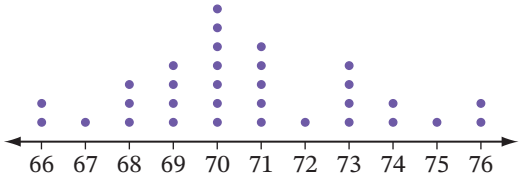
**7** For each set of data, find: 9.01

**i** the mode **ii** the mean (correct to one decimal place)

**iii** the median **iv** the range

**a** 47 46 50 51 48 48 52 46 49

**b**




**8** Classify whether each type of data is categorical or numerical. If categorical, then classify whether it is nominal or ordinal. If numerical, then classify whether it is discrete or continuous. 9.05

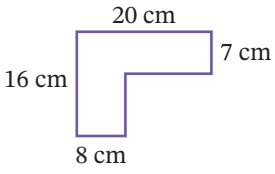
**a** the age Year 9 students turn this year **b** colour of cars in a car park

**c** travelling time between home and work **d** amount of rainfall

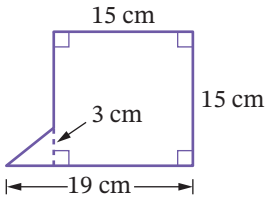
**9** For each composite shape, find: 10.04

**i** its perimeter **ii** its area

**a**




**b**




**10** Nadia's gross pay for a fortnight is $3038. She pays $724.50 in tax and her other deductions amount to $285. 8.05

**a** Calculate Nadia's net pay.

**b** Express her net pay as a percentage (correct to one decimal place) of her gross pay.

**11** For the data in this stem-and-leaf plot, which statement is true? Select the correct answer **A**, **B**, **C** or **D**. 9.02

| Stem | Leaf |
|---|---|
| 6 | 0 1 5 |
| 7 | 3 4 4 8 9 |
| 8 | 0 1 2 2 5 6 6 6 |
| 9 | 3 4 7 |
| 10 | 6 |

**A** The mean is 82.

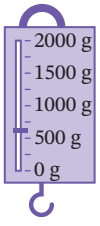
**B** The mode is 6.

**C** The range is 46.

**D** The median is 81.

**12** Sefa is paid $27.40 per hour. Calculate his earnings for working 38 hours at normal time, 8 hours at time-and-a-half and 4 hours at double time. 8.02

**13** What is the absolute error for these measuring scales? 10.02

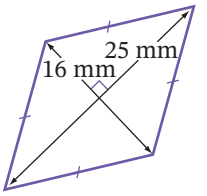



Foundation Standard Complex

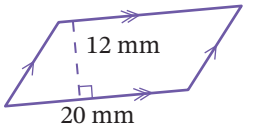
10.05

**14** Find the area of each quadrilateral.

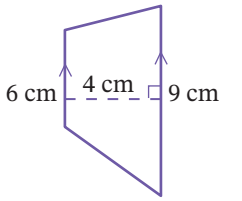
**a**




**b**




**c**




8.03

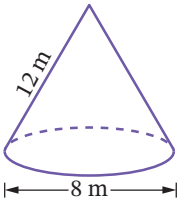
**15** Thiago earns a commission of 2.75% on every property he sells. Calculate his commission for selling an apartment for $635 200.

PATH

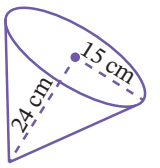
10.10

**16** Calculate the volume of each solid (correct to one decimal place where necessary).

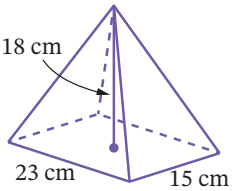
**a**



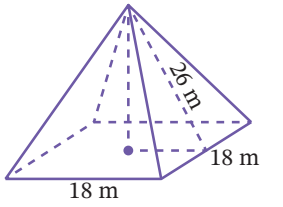

**b**




**c**




**d**



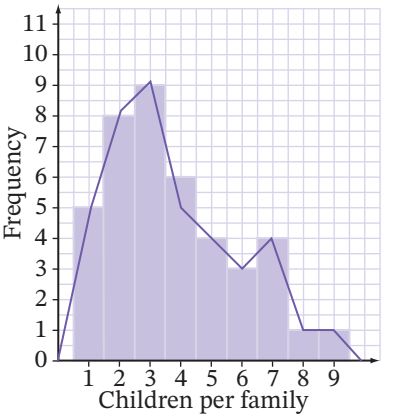

9.03

**17** A sample of students was surveyed on the number of children in their families, and the results were graphed on a frequency histogram and polygon.

**a** Describe the shape of the distribution.

**b** Where does clustering occur?

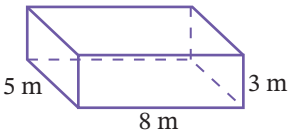
**c** Find the mode and range.



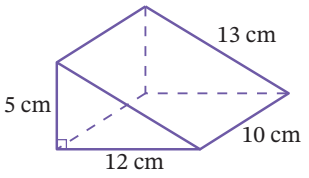

10.07

**18** Find the surface area of each prism.

**a**

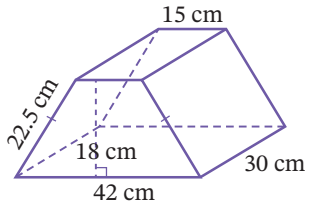



**b**




ž

**c**




10.03

**19** The diameter of a ball is 8 cm, measured to the nearest centimetre.

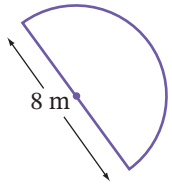
**a** What is the absolute error of this measurement?

**b** Calculate its percentage error.

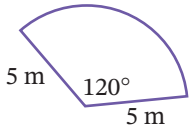
**20** For each circular shape, calculate, correct to one decimal place: 10.06

**i** its perimeter **ii** its area.

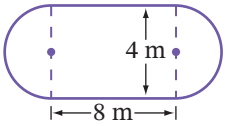
**a**




**b**




**c**



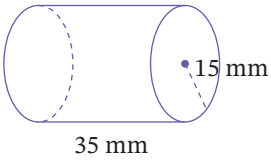

**21** In a survey, a sample of people at a park were asked 'Are you employed?'. 9.06

**a** Why might this sample of people be biased?

**b** What is wrong with the survey question?

**c** Suggest a better-worded question.

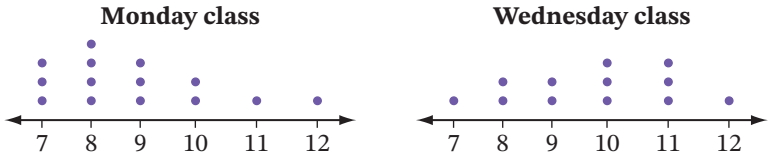
**22** Find, correct to 3 significant figures, the volume of this cylinder. 10.09




**23** Find, correct to 3 significant figures, the surface area of the cylinder above if it is open at one end. 10.08

**24** Brooke is on a salary and earns $7850 per month. Calculate Brooke's fortnightly income. 8.01

**25** These dot plots show the ages of students in 2 Taekwondo classes. 9.04




**a** Describe the shape of the age distribution for each class.

**b** Find the mean for each class.

**c** Find the median for each class.

**d** Which class had more consistent ages?

**e** Which class had more older students?

**26** A cube has a surface area of 614 $cm^2$. Find its length, correct to one decimal place. 10.07

PATH

**27** The volume of a square pyramid is 2000 $cm^3$. Calculate, correct to one decimal place, the length of the base of the pyramid if the perpendicular height is 25 cm. 10.10

**28** Celine earns $1490.40 by working 38 hours at normal rates, 4 hours at time-and-a-half and 1 hour at double time. What is Celine's normal rate of pay? 8.02

**29** Rabibah's mean mark for 4 science tests was 81%. After her 5th test, her mean test mark was 84%. What mark did Rabibah score in the 5th test? 9.01

Foundation Standard Complex